AEO «Nazarbayev intellectual schools»
Center of pedagogical measurements

Collection of students' example answers of

EXTERNAL SUMMATIVE ASSESSMENT

CHEMISTRY

Grade 12

Astana, 2018

CONTENT

External Summative Assessment Overview

Paper 2	120 minutes
All questions are compulsory. Learners answer between 8 and 12 questions, which may be broken into subparts. The questions will assess the learners' knowledge and their ability of handling, applying and evaluating information. Calculators are allowed.	
100 marks are 56% of total amount of marks	
Paper 3	**120 minutes**
All questions are compulsory. This paper consists of two or three experiments. The experiments will assess the learners' knowledge, their practical skills of planning, analysis and evaluation. Calculators are allowed.	
Marks are 22% of total amount of marks	

Grade Descriptions

Grade	Grade Description
Grade A	Demonstrates a wide and detailed knowledge of the subject with very few omissions, and has a clear understanding of the principles on which the subject is based and the manner in which it functions. The principles can be applied in both familiar and unfamiliar situations. Has a good ability to evaluate hypotheses. Answers given are well-expressed, direct and relevant and complex calculations are accurate and correctly set out. Writes accurate equations for most chemical reactions. Solves problems in situations involving a wide range of variables. Is able to generate a hypothesis to explain theories and phenomena, whilst making predictions and putting forward new hypotheses after evaluating available information effectively. Can competently design and plan investigations using suitable methods. Once completed, interprets and evaluates observations and experimental data, presenting evidence in a range of appropriate ways, can evaluate and suggest improvements to ensure precision and accuracy. From this will draw a precise set of conclusions, including next steps where necessary.
Grade C	Demonstrates a sound knowledge in many areas of the subject with some omissions, and has an understanding of many of the principles on which the subject is based and the manner in which it functions. The principles can be applied most effectively in familiar and occasionally unfamiliar situations. Has a reasonable ability to evaluate information and hypotheses. Answers given are often well-expressed, relevant and calculations frequently produce the correct answer. Writes equations for a reasonable number of chemical reactions. Solves problems involving more then one step but with a limited range of variables. Is able to generate a simple hypothesis to test a theory and make a

	prediction. Can generate a simple hypothesis to explain a given set of facts and data. Is able to plan a scientific task, such as a practical procedure, to test an idea, answer a question, or solve a problem and will present evidence in an appropriate way. From this can draw conclusions consistent with the evidence collected.
Grade E	Demonstrates a basic knowledge of the simple areas of the subject with some important omissions, and has a limited understanding of the principles on which the subject is based and the manner in which it functions. The principles are generally only applied effectively in familiar situations. Has some ability to evaluate information and hypotheses. Answers given often include relevant points but can be confused with irrelevant additions. Simple calculations are sometimes accurate but more complicated calculations have a tendency to generate error and can become unclear. Writes equations for some straightforward chemical reactions. Can solve a problem involving one step where only a minor manipulation of data is needed. Will recognise a hypothesis that explains a set of facts or data. Can plan a scientific task, such as a practical procedure, to test a basic idea, answer a simple question, or solve a straightforward problem. Can draw simple conclusions consistent with the evidence collected and present evidence as charts, tables or graphs.

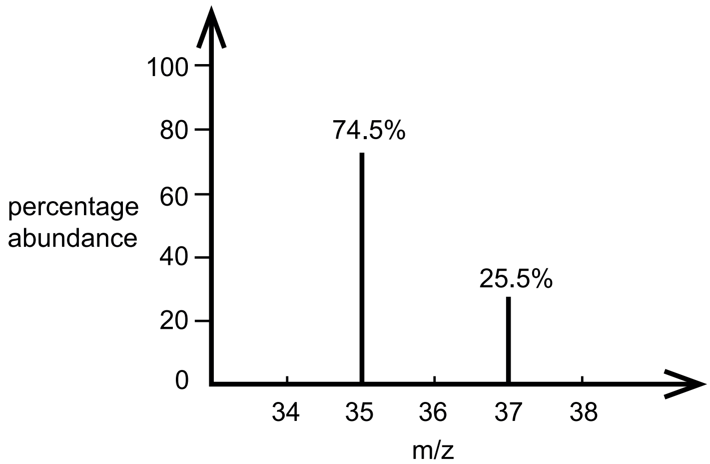
Paper 2

1(a,b,c) questions

1 Chlorine has two natural isotopes ^{35}Cl and ^{37}Cl.

The mass spectrum of a sample of chlorine is shown.

Only the peaks due to Cl^+ are included in the spectrum.



(a) Show that the relative atomic mass of chlorine is 35.5. Use information from the spectrum.

[1]

(b) Chlorine reacts with sodium to form sodium chloride.

Draw diagrams to show the full electronic configurations and charges of the ions in sodium chloride.

[3]

(c) Sodium chloride has a high melting point.

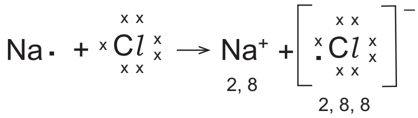
Describe the structure and bonding in sodium chloride.

Explain why the melting point is high.

....

....

.... [2]

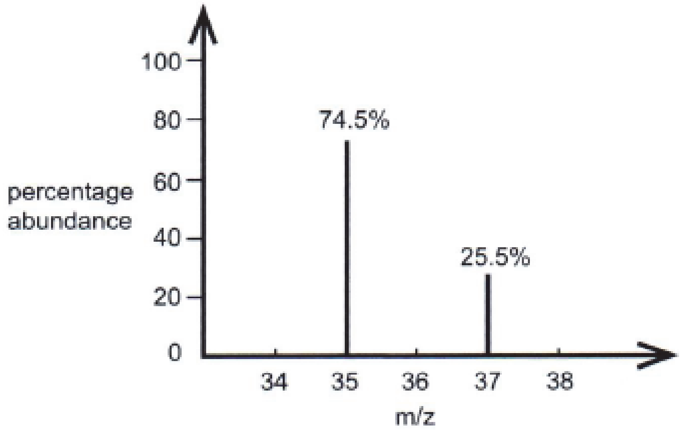
Question	Answer	Marks	Additional guidance
1 (a)	$\dfrac{74.5 \times 35 + 25.5 \times 37}{100}$	1 **[1]**	**ALLOW** any correct method
1 (b)		1 1 1 **[3]**	1 mark for the sodium ion electronic configuration **ALLOW** 2.8 1 mark for the chloride ion electronic configuration **ALLOW** 2.8.8 1 mark for both charges being correct **IGNORE** left hand side
1 (c)	Giant ionic / lattice with electrostatic attraction between ions strong attractions / lot of energy needed to break attractions	1 1 **[2]**	**ALLOW** the diagram with minimum 4 particles

Candidate A answer

1 Chlorine has two natural isotopes ^{35}Cl and ^{37}Cl.

The mass spectrum of a sample of chlorine is shown.

Only the peaks due to Cl^+ are included in the spectrum.



(a) Show that the relative atomic mass of chlorine is 35.5. Use information from the spectrum.

$$\frac{35 \cdot 74{,}5 + 25{,}5 \cdot 37}{100} = 35{,}5$$ ✓

[1]

For Examiner's Use

1

(b) Chlorine reacts with sodium to form sodium chloride.

Draw diagrams to show the full electronic configurations and charges of the ions in sodium chloride.

$\overset{+1}{Na}\overset{-1}{Cl}$

Na))) 2 8 1 $1s^2 2s^2 2p^6 3s^1$

Cl))) 2 8 7 $1s^2 2s^2 2p^6 3s^2 3p^5$

Na^{+1} $1s^2 2s^2 2p^6$

Cl^{-1} $1s^2 2s^2 2p^6 3s^2 3p^6$

[3] 3

(c) Sodium chloride has a high melting point.

Describe the structure and bonding in sodium chloride.

Explain why the melting point is high.

Structure – giant ; bonding – ionic

Consists of cation and anion

Electrostatic force is high [2] 2

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Comments for candidate A answer

1a

Candidate used information from the spectrum and completely showed correct method to calculate relative atomic mass

1b

Candidate completely drawn diagrams to show the full electronic configurations and positive and negative charges of the ions in sodium chloride.

1c

Candidate correct described the structure and bonding in sodium chloride.

Candidate completely explained why the melting point is high with appropriate key words.

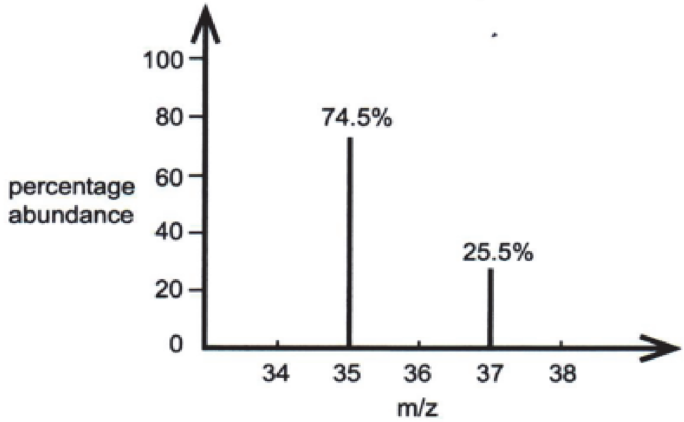
Candidate C answer

For Examiner' Use

1 Chlorine has two natural isotopes ^{35}Cl and ^{37}Cl.

The mass spectrum of a sample of chlorine is shown.

Only the peaks due to Cl^+ are included in the spectrum.



(a) Show that the relative atomic mass of chlorine is 35.5. Use information from the spectrum.

$$\frac{35 \cdot 74{,}5 + 37 \cdot 25{,}5}{100} = 35{,}5$$

[1] 1

(b) Chlorine reacts with sodium to form sodium chloride.

Draw diagrams to show the full electronic configurations and charges of the ions in sodium chloride.

Na^+ $1s^2 2s^2 2p^6$

Cl^- $1s^2 2s^2 2p^6 3s^2 3p^6$

[3] 3

(c) Sodium chloride has a high melting point.

Describe the structure and bonding in sodium chloride.

Explain why the melting point is high.

Sodium chloride has ionic bond and giant ionic structure, therefore melting point of NaCl is high [2] 1

Comments for candidate C answer

1a

Candidate used information from the spectrum and completely showed correct method to calculate relative atomic mass

1b

Candidate completely drawn diagrams to show the full electronic configurations and positive, negative charges of the ions in sodium chloride.

1c

Candidate correct described the structure and bonding in sodium chloride.

Candidate no arguments for explanation why the melting point is high

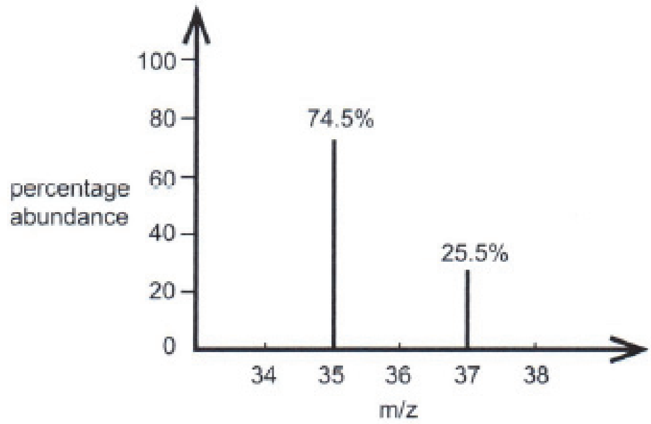
Candidate E answer

For Examiner Use

1 Chlorine has two natural isotopes ^{35}Cl and ^{37}Cl.

The mass spectrum of a sample of chlorine is shown.

Only the peaks due to Cl^+ are included in the spectrum.



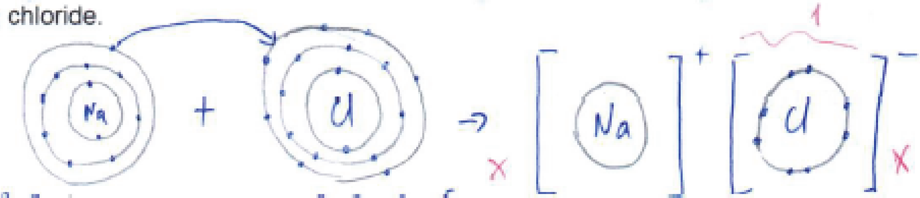
(a) Show that the relative atomic mass of chlorine is 35.5. Use information from the spectrum.

$A_r = \frac{35\times74.5\% + 25.5\%\times37}{100\%} = 35.5$ ✓

[1] 1

(b) Chlorine reacts with sodium to form sodium chloride.

Draw diagrams to show the full electronic configurations and charges of the ions in sodium chloride.



$1s^2 2s^2 2p^6 3s^1$ $1s^2 2s^2 2p^6 3s^2 3p^5$. [Ne] [Ar] [3] 1

(c) Sodium chloride has a high melting point.

Describe the structure and bonding in sodium chloride.

Explain why the melting point is high.

Because there is an Ionic Bonding × between sodium and chlorine ions that requires a lot ✓ of energy to break it down, that is why mp is high [2] 1

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Comments for candidate E answer

1a

Candidate used information from the spectrum and completely showed correct method to calculate relative atomic mass

1b

Candidate completely drawn diagrams to show the full electronic configurations and charges of the ions in sodium chloride.

1c

Candidate showed ion charges, but didn't draw the full electron configuration of ions

1(d) question

(d) A Born-Haber cycle shows the enthalpy changes associated with the formation of ionic compounds.

The table shows some of these enthalpy changes.

process	**enthalpy change / kJ mol^{-1}**
$Na(s) \rightarrow Na(g)$	+108
$Na(g) \rightarrow Na^+(g) + e^-$	+495
$\frac{1}{2} Cl_2(g) \rightarrow Cl(g)$	+121
$Cl(g) \rightarrow Cl^-(g)$	–349
$Na(s) + \frac{1}{2} Cl_2(g) \rightarrow NaCl(s)$	–410

The lattice energy for sodium chloride is the enthalpy change associated with the process shown in the equation.

$$Na^+(g) + Cl^-(g) \rightarrow NaCl(s)$$

Calculate the lattice energy for sodium chloride.

Use data from the table.

1 (d)	For a correct cycle $\Delta H = (-410) - (108 + 495 + 121 - 349)$ answer = –785 (kJ mol-1)	1 1 1 **[3]**	this expression is worth 2 marks –785 = 3/3 +785 = 2/3 –677 = 2/3; +677 = 1/3 –375 = 2/3; +375 = 1/3 –290 = 2/3; +290 = 1/3 –664 = 2/3; +664 = 1/3 –1134 = 2/3; +1134 = 1/3

lattice energy = ... kJ mol^{-1} [3]

Candidate A answer

(d) A Born-Haber cycle shows the enthalpy changes associated with the formation of ionic compounds.

For Examiner's Use

The table shows some of these enthalpy changes.

process	enthalpy change / kJ mol^{-1}
$Na(s) \rightarrow Na(g)$	+108
$Na(g) \rightarrow Na^+(g) + e^-$	+495
$\frac{1}{2}Cl_2(g) \rightarrow Cl(g)$	+121
$Cl(g) \rightarrow Cl^-(g)$	–349
$Na(s) + \frac{1}{2}Cl_2(g) \rightarrow NaCl(s)$	–410

The lattice energy for sodium chloride is the enthalpy change associated with the process shown in the equation.

$$Na^+(g) + Cl^-(g) \rightarrow NaCl(s)$$

Calculate the lattice energy for sodium chloride.

Use data from the table.

$\frac{1}{2} Cl_2g \rightarrow Clg$

$Na_g \rightarrow Na^+$

$Na_s \rightarrow Na_g$

NaCl

formation

$Cl(g) \rightarrow Cl^-(g)$

lattise

$Na_s + \frac{1}{2} Cl_2 \rightarrow NaCl$

$H_{lattice} = H_f - H_{processes}$ ✓ ✓

$H_{lattice} = -410 - (108 + 495 + 121 - 349) =$ ✓ ✓

$= -785$ kJ $mole^{-1}$ ✓ ✓

lattice energy = -785 kJ mol^{-1} [3]

3 3

Comments for candidate A answer

1d

Candidate did correct expression for calculation the lattice energy for sodium chloride using data from the table

Candidate C answer

(d) A Born-Haber cycle shows the enthalpy changes associated with the formation of ionic compounds.

For Examiner's Use

The table shows some of these enthalpy changes.

process	enthalpy change / kJ mol^{-1}
$Na(s) \rightarrow Na(g)$	+108
$Na(g) \rightarrow Na^+(g) + e^-$	+495
$\frac{1}{2}Cl_2(g) \rightarrow Cl(g)$	+121
$Cl(g) \rightarrow Cl^-(g)$	–349
$Na(s) + \frac{1}{2}Cl_2(g) \rightarrow NaCl(s)$	–410

The lattice energy for sodium chloride is the enthalpy change associated with the process shown in the equation.

$$Na^+(g) + Cl^-(g) \rightarrow NaCl(s)$$

Calculate the lattice energy for sodium chloride.

Use data from the table.

$E_f = E_a + \frac{1}{2}E_{(cl)} + E_i + E_{e.a.} + E_L$

$E_L = E_a + \frac{1}{2}E_{(cl)} + E_i + E_{e.a} - E_f$

$E_L = 108 + 495 + 121 + 349 + 410 = 785$

$102 + 495 + \frac{121}{2} - 349 + 410 = 724,5$

lattice energy = 785 kJ mol^{-1} [3] 2 2

Comments for candidate C answer

1d

Candidate did incorrect expression for calculation the lattice energy for sodium chloride using data from the table. The candidate gets the calculated value correctly with the wrong sign and therefore gets 2 points

Candidate E answer

(d) A Born-Haber cycle shows the enthalpy changes associated with the formation of ionic compounds.

For Examiner's Use

The table shows some of these enthalpy changes.

process	enthalpy change / kJ mol^{-1}
$Na(s) \rightarrow Na(g)$	+108
$Na(g) \rightarrow Na^{+}(g) + e^{-}$	+495
$\frac{1}{2}Cl_2(g) \rightarrow Cl(g)$	+121
$Cl(g) \rightarrow Cl^{-}(g)$	–349
$Na(s) + \frac{1}{2}Cl_2(g) \rightarrow NaCl(s)$	–410

The lattice energy for sodium chloride is the enthalpy change associated with the process shown in the equation.

$$Na^{+}(g) + \overset{349}{Cl^{-}(g)} \rightarrow \overset{-410}{NaCl(s)}$$

Calculate the lattice energy for sodium chloride.

Use data from the table.

(d) A Born-Haber cycle shows the enthalpy changes associated with the formation of ionic compounds.

For Examiner's Use

The table shows some of these enthalpy changes.

process	enthalpy change / kJ mol^{-1}
$Na(s) \rightarrow Na(g)$	+108
$Na(g) \rightarrow Na^{+}(g) + e^{-}$	+495
$\frac{1}{2}Cl_2(g) \rightarrow Cl(g)$	+121
$Cl(g) \rightarrow Cl^{-}(g)$	–349
$Na(s) + \frac{1}{2}Cl_2(g) \rightarrow NaCl(s)$	–410

The lattice energy for sodium chloride is the enthalpy change associated with the process shown in the equation.

$$Na^{+}(g) + Cl^{-}(g) \rightarrow NaCl(s)$$

Calculate the lattice energy for sodium chloride.

Use data from the table.

$\Delta H_{latt} = \Delta H_{at} + \Delta H_{i.e.} + \Delta H_{e.a.} + \Delta H_{f} = 108 + 495$

$+ 121 - 349 - 410 = -35$

lattice energy = –35 XXX kJ mol^{-1} [3]

0

Comments for candidate E answer

1d

Candidate did incorrect expression for calculation the lattice energy for sodium chloride using data from the table. The candidate calculated value with incorrect expression and therefore gets 0 points

2(a,b) questions

2 **(a)** State **and** explain the trend in melting point down Group 2 from Mg to Ba.

....

....

....

....

.... [3]

(b) Write balanced equations for the neutralisation reactions of magnesium oxide and magnesium hydroxide with hydrochloric acid.

equation with magnesium oxide

....

equation with magnesium hydroxide

....

[2]

2 (a)	decrease radius/size of ions (atoms) /distance between nucleous and outer shell electrons increasing / charge density decreasing strength of attraction (for delocalised electrons) decreases	1 1 1 **[3]**	**ALLOW** melting point ↓ **ALLOW** strength of the metallic bond decreases
2 (b)	$MgO + 2HCl \rightarrow MgCl_2 + H_2O$ $Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$	1 1 **[2]**	**ALLOW** multiples **ALLOW** multiples

Candidate A answer

2 (a) State **and** explain the trend in melting point down Group 2 from Mg to Ba.

Melting point decreases down the group. Metals have metallic lattice made of cations and delocalized electrons. Down the group size of the cations increase due to increasing of shells numbers. This leads to the decreasing of attraction between cations and delocalized sea, because distance from nucleus to outer shell e^- increase and cation cannot attract sea electrons stronger. [3] 3

(b) Write balanced equations for the neutralisation reactions of magnesium oxide and magnesium hydroxide with hydrochloric acid.

equation with magnesium oxide

$MgO_{(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_2O_{(l)}$

equation with magnesium hydroxide

$Mg(OH)_{2(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + 2H_2O_{(l)}$

[2] 2

For Examiner's Use

Comments for candidate A answer

2a

Candidate correct define the trend in melting point down Group 2 from Mg to Ba. Candidate explained the trend in melting point down Group 2 in point of atomic structure and sense it effect for physical properties (melting point)

2b

Candidate correct wrote balanced equations for the neutralisation reactions of magnesium oxide and magnesium hydroxide with hydrochloric acid.

Candidate C answer

2 **(a)** State **and** explain the trend in melting point down Group 2 from Mg to Ba.

For Examiner's Use

Melting point down the group 2 from Mg to Ba decreases, because of shielding effect, as nuclei is being heavier, number of shells increases and the distance between nucleus and shells is big, that is why it is easy to break it

[3] 2

(b) Write balanced equations for the neutralisation reactions of magnesium oxide and magnesium hydroxide with hydrochloric acid.

equation with magnesium oxide

$MgO_{(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_2O_{(l)}$

equation with magnesium hydroxide

$Mg(OH)_{2(aq)} + 2HCl_{(aq)} \rightarrow MgCl_{2(l)} + 2H_2O_{(l)}$

[2] 2

Comments for candidate C answer

2a

Candidate correct define the trend in melting point down Group 2 from Mg to Ba. Candidate explained the trend in melting point down Group 2 in point of atomic structure but could't determine it effect for physical properties (melting point)

2b

Candidate correct wrote balanced equations for the neutralisation reactions of magnesium oxide and magnesium hydroxide with hydrochloric acid.

Candidate E answer

2 (a) State **and** explain the trend in melting point down Group 2 from Mg to Ba.

For Examiner's Use

Melting point of Group 2 is increase down the group, because with increasing the radius of atom there it require more energy to melting

[3] 1

(b) Write balanced equations for the neutralisation reactions of magnesium oxide and magnesium hydroxide with hydrochloric acid.

equation with magnesium oxide

$MgO_{(aq)} + 2HCl_{(aq)} \rightarrow MgCl_2 + H_2O_{(l)}$

equation with magnesium hydroxide

$Mg(OH)_{2(aq)} + 2HCl_{(aq)} \rightarrow MgCl_{2(s)} + 2H_2O_{(l)}$

[2] 2

Comments for candidate E answer

2a Candidate incorrect define the trend in melting point down Group 2 from Mg to Ba. Candidate explained the trend in melting point down Group 2 in point of atomic structure but could't determine it effect for physical properties (melting point)

2b

Candidate correct wrote both equations and balanced them

2 (c) question

(c) Barium sulfate is an insoluble salt.

(i) State one use in medicine for barium sulfate.

.... [1]

(ii) A technician mixes $1.0\,dm^3$ of $0.25\,mol\,dm^{-3}$ $BaCl_2$ with $1.0\,dm^3$ of $0.50\,mol\,dm^{-3}$ K_2SO_4.

A precipitation reaction takes place.

$$BaCl_2(aq) + K_2SO_4(aq) \rightarrow BaSO_4(s) + 2KCl(aq)$$

Calculate the concentration of each aqueous ion left in the solution after the $BaSO_4$ precipitate is formed.

$[Ba^{2+}(aq)]$ = mol dm^{-3}

$[Cl^-(aq)]$ = mol dm^{-3}

$[K^+(aq)]$ = mol dm^{-3}

$[SO_4^{2-}(aq)]$ = ... mol dm^{-3} [4]

2 (c) (i)	To prepare 'barium meal' for radiography diagnostics / X-ray analysis	1 **[1]**	

2 (c) (ii)	(moles of Ba^{2+} to start with = 0.25 all of which reacts so) $[Ba^{2+}]$ = 0 (mol dm^{-3})	1	if every concentration is doubled then award 3 marks for the calculations
	(moles of Cl^- to start with = 0.50 so moles of Cl^- at end = 0.50 and) $[Cl^-]$ = (0.25 moldm^{-3})	1	
	(moles of K^+ to start with and at the end = 1.00 so) $[K^+]$ = 0.50 (mol dm^{-3})	1	if concentrations are incorrect allow 1 mark for correct moles at end for two ions and 2 marks for all four ions
	(moles of SO_4^{2-} to start with = 0.50 so moles at end = 0.25 and) $[SO_4^{2-}]$ = 0.125 (mol dm^{-3})	1 **[4]**	

Candidate A answer

(c) Barium sulfate is an insoluble salt.

For Examiner's Use

(i) State one use in medicine for barium sulfate.

In X-ray observation, Rentgen ✓ [1] 1 1

(ii) A technician mixes 1.0 dm^3 of 0.25 mol dm^{-3} $BaCl_2$ with 1.0 dm^3 of 0.50 mol dm^{-3} K_2SO_4.

A precipitation reaction takes place.

$$\overset{0.25}{BaCl_2(aq)} + K_2SO_4(aq) \rightarrow BaSO_4(s) + \overset{x}{2KCl(aq)}$$

1 2

Calculate the concentration of each aqueous ion left in the solution after the $BaSO_4$ precipitate is formed.

$n(BaCl_2) = CV = 0.25 \cdot 1 = 0.25$ mole

$n(K_2SO_4) = CV = 0.5 \cdot 1 = 0.5$ mole

$C = \frac{n}{V}$

$n(KCl) = \underline{0.5 \text{ mole}}$

$n(K_2SO_4)_{left} = 0.5 - 0.25 = \underline{0.25}$

$C(KCl) = \frac{0.5}{2} = 0.25 \Rightarrow [K^+] = 0.25 \text{ mol dm}^{-3}$ $[Cl^-] = 0.25 \text{ mol dm}^{-3}$

$C(K_2SO_4) = \frac{0.25}{2} = 0.125$

$\Downarrow$

$2[K^+] = 0.25$

$[SO_4^{2-}] = 0.125$

$[Ba^{2+}(aq)] =$ 0 ✓ mol dm^{-3}

$[Cl^-(aq)] =$ 0.25 ✓ mol dm^{-3}

$[K^+(aq)] =$ 0.5 ✓ mol dm^{-3}

$[SO_4^{2-}(aq)] =$ 0.125 ✓ mol dm^{-3} [4]

4

Comments for candidate A answer

2c (i) Candidate correct defined usage barium sulfate in medicine

2c (ii) Candidate correct calculated the concentration of each aqueous ion left in the solution after the BaSO4 precipitate is formed.

Candidate C answer

(c) Barium sulfate is an insoluble salt.

(i) State one use in medicine for barium sulfate.

Barium meal in X-ray (opaque for X-ray) in gastroenterology [1] 1

(ii) A technician mixes 1.0 dm^3 of 0.25 mol dm^{-3} $BaCl_2$ with 1.0 dm^3 of 0.50 mol dm^{-3} K_2SO_4.

A precipitation reaction takes place.

$$\overset{0.2}{BaCl_2(aq)} + K_2SO_4(aq) \rightarrow BaSO_4(s) + 2KCl(aq)$$

Calculate the concentration of each aqueous ion left in the solution after the $BaSO_4$ precipitate is formed.

$$Ba^{2+}_{(aq)} + 2Cl^-_{(aq)} + 2K^+_{(aq)} + SO_4^{2-}{}_{(aq)} \rightarrow BaSO_{4(s)} + 2K^+_{(aq)} + 2Cl^-_{(aq)}$$

$$Ba^{2+}{}_{(aq)} + SO_4^{2-}{}_{(aq)} \rightarrow BaSO_{4(s)}$$

$n(BaCl_2) = 1 \times 0.25 = 0.25 mol$ $[Ba^{2+}] = \frac{0.25 mol}{2 dm^3} = 0.125 mol dm^{-3}$

$n(K_2SO_4) = 1 \times 0.5 = 0.5 mol$

$n(BaSO_4) = 0.25 mol$

$n(KCl) = 0.25 \times 2 = 0.5 mol$ $[Cl^-] = \frac{0.5 mol}{2 dm^3} = 0.250 mol dm^{-3}$

$[Ba^{2+}(aq)]$ = 0.125 ✗ mol dm^{-3}

$[Cl^-(aq)]$ = 0.250 ✓ mol dm^{-3}

$[K^+(aq)]$ = 0.250 ✗ mol dm^{-3}

$[SO_4^{2-}(aq)]$ = 0.125 ✓ mol dm^{-3} [4] 2

For Examiner's Use

Comments for candidate C answer

2c (i) Candidate correct defined usage barium sulfate in medicine

2c (ii) Candidate correct calculated the initiatial concentration of each aqueous ion in the solution, but doesn't calculate concentration of aqueous ion which reacts each other and after the $BaSO_4$ precipitate is formed.

Candidate E answer

(c) Barium sulfate is an insoluble salt.

(i) State one use in medicine for barium sulfate.

~~Drehstahl~~ ~~Painkiller~~, absorbent ✗ .... [1]

0

(ii) A technician mixes 1.0 dm^3 of 0.25 mol dm^{-3} $BaCl_2$ with 1.0 dm^3 of 0.50 mol dm^{-3} K_2SO_4.

A precipitation reaction takes place.

$$BaCl_2(aq) + K_2SO_4(aq) \rightarrow BaSO_4(s) + 2KCl(aq)$$

Calculate the concentration of each aqueous ion left in the solution after the $BaSO_4$ precipitate is formed.

$$Ba^{2+}_{(aq)} + 2Cl^-_{(aq)} + 2K^+_{(aq)} + SO_4^{2-}{}_{(aq)} \rightarrow BaSO_{4(s)} + 2K^+_{(aq)} + 2Cl^-_{(aq)}$$

$$Ba^{2+}_{(aq)} + SO_4^{2-}{}_{(aq)} \rightarrow BaSO_{4(s)}$$

$n(BaCl_2) = 1 \times 0.25 = 0.25 mol$ $[Ba^{2+}] = \frac{0.25 mol}{2 dm^3} = 0.125 mol dm^{-3}$

$n(K_2SO_4) = 1 \times 0.5 = 0.5 mol$

$n(BaSO_4) = 0.25 mol$

$n(KCl) = 0.25 \times 2 = 0.5 mol$ $[Cl^-] = \frac{0.5 mol}{2 dm^3} = 0.250 mol dm^{-3}$

$[Ba^{2+}(aq)]$ = 0.125 ✗ mol dm^{-3}

$[Cl^-(aq)]$ = 0.250 ✓ mol dm^{-3}

$[K^+(aq)]$ = 0.250 ✗ mol dm^{-3}

$[SO_4^{2-}(aq)]$ = 0.125 ✓ mol dm^{-3} [4]

2

Comments for candidate E answer

2c (i) Candidate incorrect defined usage barium sulfate in medicine

2c (ii) Candidate correct calculated the initiatial concentration of each aqueous ion in the solution, but doesn't calculate concentration of aqueous ion which reacts each other and after the $BaSO_4$ precipitate is formed.

3(a,b questions)

3 **(a)** One physical property of transition elements is that they are good thermal conductors.

State **two other** physical properties of transition elements.

....

.... [1]

(b) **(i)** Complete the full electronic configuration for Cr.

$1s^2$ .... [1]

(ii) State the oxidation state of chromium in each of these compounds.

$Cr_2(SO_4)_3$ ....

K_2CrO_4 .... [1]

3 (a)	Any 2 from: high melting/boiling point high density high strength/hard good conductors of electricity malleable ductile	1 **[1]**	
3 (b) (i)	$(1s^2)2s^22p^63s^23p^63d^54s^1$	1 **[1]**	ALLOW ….$4s^13d^5$
3 (b) (ii)	$Cr_2(SO_4)_3$ = (+)3	1	

	and K_2CrO_4 = (+)6	**[1]**	

Candidate A answer

6

3 (a) One physical property of transition elements is that they are good thermal conductors.

State **two other** physical properties of transition elements.

high melting point

good electricity conductors [1]

For Examiner's Use

1

(b) (i) Complete the full electronic configuration for Cr.

$1s^2$ $2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$ [1]

1

(ii) State the oxidation state of chromium in each of these compounds.

$Cr_2(SO_4)_3$ Cr^{3+}

$K_2\overset{+6}{Cr}O_4^{-2}$ Cr^{6+} [1]

1

Comments for candidate A answer

3a

Candidate correct defined two physical properties of transition elements.

3b(i)

Candidate correct complete the electronic configuration for chromium

3b(ii)

Candidate correct defined the oxidation state of chromium in each of given compounds.

Candidate C answer

3 **(a)** One physical property of transition elements is that they are good thermal conductors.

State **two other** physical properties of transition elements.

conduct electricity, rigid, ~~can form~~ hard ✓✓

[1] 1 1

(b) **(i)** Complete the full electronic configuration for Cr. 24 2 8 8 6

$1s^2$ $2s^2 2p^6 3s^2 3p^6 4s^2 4p^4$ ✗ ✗ [1] 0 0

(ii) State the oxidation state of chromium in each of these compounds.

$Cr_2(SO_4)_3$ +3 ✓

$K_2(CrO_4)^{2-}$ +6 ✓ 1

[1] 1 1

For Examiner's Use

Comments for candidate C answer

3a

Candidate correct defined two physical properties of transition elements.

3b(i)

Candidate incorrect complete the electronic configuration for chromium

3b(ii)

Candidate correct defined the oxidation state of chromium in each of given compounds.

Candidate E answer

For Examiner's Use

3 **(a)** One physical property of transition elements is that they are good thermal conductors.

State **two other** physical properties of transition elements.

1) Transition elements have a colour X

2) Transition elements are good ~~electron~~ electricity conductors ✓ [1] 0

3) they have different oxidation states

(b) **(i)** Complete the full electronic configuration for Cr.

$1s^2$ $2s^2 2p^6 3s^2 3p^6 3d^6 4s^0$ X [1] 0

3d > 4s ↑↓ ↑↓

(ii) State the oxidation state of chromium in each of these compounds.

$Cr_2(SO_4)_3^{2-}$ $2x - 6 = 0$ $2x = 6$ $x = 3$ Cr^{+3} ✓

$K_2^+CrO_4^{2-}$ $2 + x - 8 = 0$ $x = 6$ Cr^{+6} ✓ 1 [1] 1

Comments for candidate E answer

3a

Candidate correct defined one physical properties of transition elements and due to this lost 1 point

3b(i)

Candidate incorrect complete the electronic configuration for chromium

3b(ii)

Candidate correct defined the oxidation state of chromium in each of given compounds.

3c (i,ii) question

(c) Reinecke's salt contains the complex ion $[Cr(NH_3)_2(CNS)_4]^-$

(i) Name the shape of this complex ion.

State the bond angle in this complex ion.

Draw a diagram to show the shape of this complex ion.

name of shape ....

bond angle ....

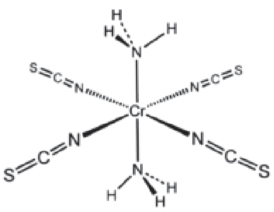
[3]

(ii) NH_3 is a monodentate ligand.

State the meaning of the term *polydentate ligand*.

....

.... [1]

3 (c) (i)	octahedral 90° 	1 1 1 **[3]**	**ALLOW** cis **IGNORE** connectivity – mark is for the shape The diagram must show 3D
3 (c) (ii)	(a polydentate ligand forms) more than one dative covalent bond with a central metal atom	1 **[1]**	

Candidate A answer

(c) Reinecke's salt contains the complex ion $[Cr(NH_3)_2(CNS)_4]^-$

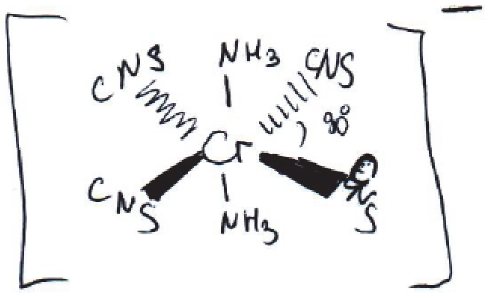
(i) Name the shape of this complex ion.

State the bond angle in this complex ion.

Draw a diagram to show the shape of this complex ion.

name of shape octahedral ✓

bond angle 90° ✓



✓

[3] 3

(ii) NH_3 is a monodentate ligand.

State the meaning of the term *polydentate ligand*.

Substance that forms more than 1 coordinate bond with central metal ion, have more than 1 el. pair. ✓ [1] 1

Comments for candidate A answer

3c(i)

Candidate correct named the shape and defined bond angle of this complex ion. The candidate correct drawn the 3D shape of this complex ion.

3c(ii)

Candidate correct defined the meaning of the term *polydentate ligand* with appropriate key words

Candidate C answer

(c) Reinecke's salt contains the complex ion $[Cr(NH_3)_2(CNS)_4]^-$

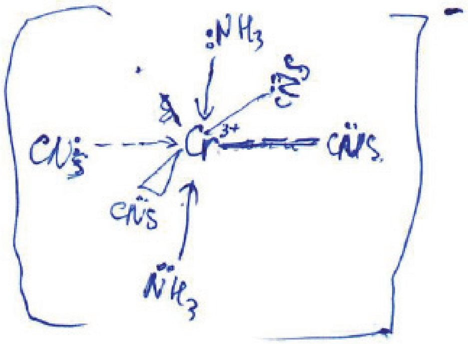
(i) Name the shape of this complex ion.

State the bond angle in this complex ion.

Draw a diagram to show the shape of this complex ion.

name of shape ....octahedral........

bond angle ....90°........



[3] 2

(ii) NH_3 is a monodentate ligand.

State the meaning of the term *polydentate ligand*.

It is ligand that has two or more dative bond with central transition metal ion. [1] 1

Comments for candidate C answer

3c(i)

Candidate correct named the shape and defined bond angle of this complex ion. The candidate incorrect drawn the 3D shape of this complex ion and lost 1 point

3c(ii)

Candidate correct defined the meaning of the term *polydentate ligand* with appropriate key words

Candidate E answer

(c) Reinecke's salt contains the complex ion $[Cr(NH_3)_2(CNS)_4]^-$

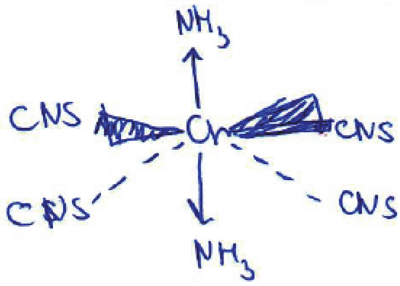
(i) Name the shape of this complex ion.

State the bond angle in this complex ion.

Draw a diagram to show the shape of this complex ion.

name of shape Octahedral

bond angle 90°



[3] 3 3

(ii) NH_3 is a monodentate ligand.

State the meaning of the term *polydentate ligand*.

....

.... [1] 0 0

Comments for candidate E answer

3c(i)

Candidate correct named the shape and defined bond angle of this complex ion. The candidate correct drawn the 3D shape of this complex ion

3c(ii)

Candidate incorrect defined the meaning of the term *polydentate ligand*

3c (iii) question

(iii) The complex ion in Reinecke's salt is deep-red in colour.

Explain why the complex ion is coloured.

....

....

....

....

.... [3]

3 (c) (iii)	(Ligands cause a) split in energies of 3d orbitals / 3d-3d transitions are possible energy / (visible) light absorbed to promote electrons colour seen is the complementary colour / light absorbed at blue/green-blue end of spectrum	1 1 1 **[3]**	

Candidate A answer

7

(iii) The complex ion in Reinecke's salt is deep-red in colour.

Explain why the complex ion is coloured.

For Examiner's Use

When complex is formed electrons in d orbital become non-degenerated. d orbital split into 2 sets, in octahedral complexes 3 in low, 2 in high energy level. When light is applied electrons in lower energy set are promoted to higher level by absorbing light energy from photons. this absorbed energy is ΔE and (~~[illegible]~~) allows colour change, complementary colour is reflected. [3]

3

Comments for candidate A answer

3c(iii)

Candidate completely explaned why the complex ion is colored with correct key words

Candidate C answer

7

(iii) The complex ion in Reinecke's salt is deep-red in colour.

Explain why the complex ion is coloured.

For Examiner's Use

When light is absorbed ~~non~~-degenerate d-orbitals split into 2 different energy levels - non-degenerate d-orbitals. Electron jumps from lower ~~one~~ energy level to higher energy level. Yellow-green color was absorbed. We see reflected colors [3]

2

Comments for candidate C answer

3c(iii)

Candidate explaned why the complex ion is colored with correct key words. If candidate would use for explanation of last stage appropriate key words would got 1 point

Candidate E answer

7

(iii) The complex ion in Reinecke's salt is deep-red in colour. | *For Examiner's Use*

Explain why the complex ion is coloured.

It has such colour due to ligands, so they form different energy gaps and therefore molecule absorb colour of certain wavelength

[3] 1 1

Comments for candidate E answer

3c(iii)

Candidate incompletely explaned why the complex ion is colored , for which lost 2 points

4 a,b questions

4 Halogenoalkanes are a large class of organic compounds.

(a) Draw all of the structural formula of isomers with the molecular formula C_4H_9Cl.

[2]

(b) Halogenoalkanes undergo reactions with nucleophiles.

(i) Define the term *nucleophile* and give **two** examples of nucleophiles that react with halogenoalkanes.

....

....

.... [2]

(ii) State why halogenoalkane molecules undergo reactions with nucleophiles.

....

.... [1]

4 (a)	$CH_2 - CH_2 - CH_2 - CH_3$ | Cl $CH_3 - CH - CH_2 - CH_3$ | Cl CH_3 | $CH_2 - CH - CH_3$ | Cl CH_3 | $CH_3 - C - CH_3$ | Cl	**[max2]**	4 isomers correct = 2/2 2 or 3 isomers correct = 1/2 **ALLOW** any structure e.g. displayed, skeletal or structural providing that it is unambiguous **ALLOW** if the chlorine replaced by the another halogen **IGNORE** any names given

4 (b) (i)	lone / electron <u>pair</u> donor Any **two** from: OH^- H_2O NH_3 / NH_2^- amines CN^-	1 1 **[2]**	**ALLOW** correct names or formulae but if both given they must match Consider at coordination meeting other nucleophiles, e.g. halide ions
4 (b) (ii)	C is δ+ / C–X bond is polar	1 **[1]**	

Candidate A answer

8

4 Halogenoalkanes are a large class of organic compounds.

(a) Draw all of the structural formula of isomers with the molecular formula C_4H_9Cl.

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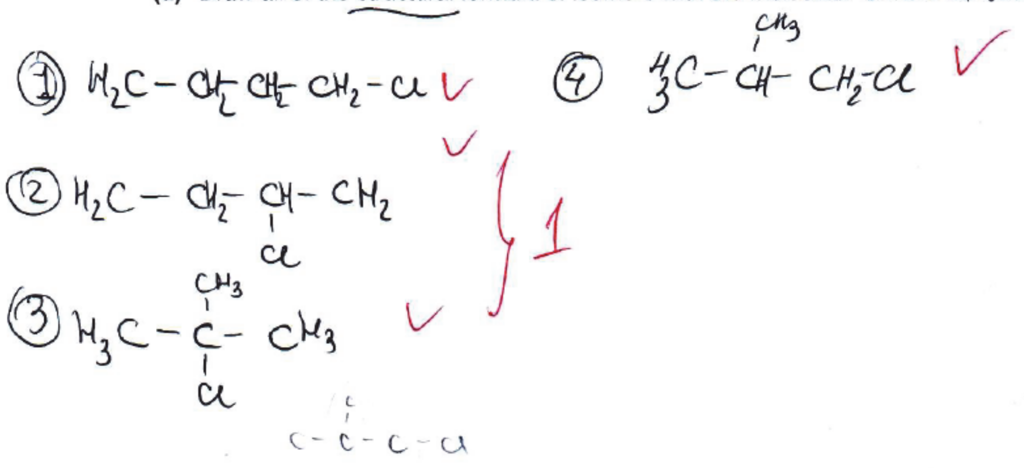


[2]

2

(b) Halogenoalkanes undergo reactions with nucleophiles.

(i) Define the term *nucleophile* and give **two** examples of nucleophiles that react with halogenoalkanes.

Nucleophiles are electron pair donors, attract to ~~E~~ positive charges. Ex: $:OH^-$; $:CN^-$

[2] 2

(ii) State why halogenoalkane molecules undergo reactions with nucleophiles.

Carbon joined to halogen is partially positive due to different electronegativities, bond is polar. [1] 1

Comments for candidate A answer

4a

Candidate correct drawn all of the possible structural formula of isomers with the molecular formula C4H9Cl. For each 2 isomers the candidate got 1 point

4b (i)

Candidate gave a correct definition with appropriate keywords for the term nucleophile. The candidate gave two correct examples of nucleophiles that react with halogenoalkanes and got full mark.

4b(ii)

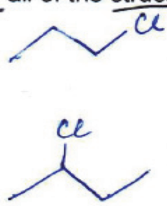
Candidate correct defined why halogenoalkane molecules undergo reactions with nucleophiles in point of view charge of halogen in halogenoalkane

Candidate C answer

4 Halogenoalkanes are a large class of organic compounds.

For Examiner's Use

(a) Draw all of the structural formula of isomers with the molecular formula C_4H_9Cl.



1

[2] 1

(b) Halogenoalkanes undergo reactions with nucleophiles.

(i) Define the term *nucleophile* and give **two** examples of nucleophiles that react with halogenoalkanes.

Nucleophile – ~~molecules~~ negative charged ~~atoms~~ which donates pair of electrons. For ex.: CN^-, OH^-

[2] 2

(ii) State why halogenoalkane molecules undergo reactions with nucleophiles.

They are polar. There is a difference in electronegativity of halogen and carbon atom. [1] 1

Comments for candidate C answer

4a

Candidate drawn 2 correct structural formula of isomers with the molecular formula C4H9CI. For which this got 1 point

4b (i)

Candidate gave a correct definition with appropriate keywords for the term nucleophile. The candidate gave two correct examples of nucleophiles that react with halogenoalkanes and got full mark.

4b(ii)

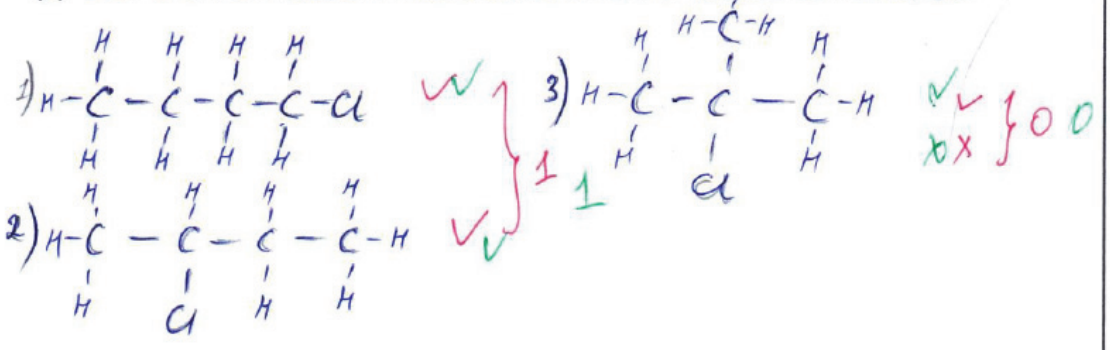
Candidate correct defined why halogenoalkane molecules undergo reactions with nucleophiles in point of view polarity of bond

Candidate E answer

4 Halogenoalkanes are a large class of organic compounds.

For Examiner's Use

(a) Draw all of the structural formula of isomers with the molecular formula C_4H_9Cl.



[2] 1

(b) Halogenoalkanes undergo reactions with nucleophiles.

(i) Define the term *nucleophile* and give **two** examples of nucleophiles that react with halogenoalkanes.

nucleophile is electron rich (lone pair) substace or ion that attracted to the compound which is electron deficient. Ex: OH^-, CN^- [2] 2

(ii) State why halogenoalkane molecules undergo reactions with nucleophiles.

because they are electron deficient that's why undergo reaction with nucleophiles [1] 0

Comments for candidate E answer

4a

Candidate drawn 3 correct structural formula of isomers with the molecular formula C4H9CI. For each 2 isomers the candidate must get 1 point, therefore candidate got 1 point

4b (i)

Candidate gave a correct definition with appropriate keywords for the term nucleophile. The candidate gave two correct examples of nucleophiles that react with halogenoalkanes and got full mark.

4b(ii)

Candidate incorrect defined why halogenoalkane molecules undergo reactions with nucleophiles in terms of electrofiles which opposite to halogen acting in halogenalkane, that is why the candidate could't got 1 point

4c,d questions

(c) Propan-1-ol is used to synthesise 1-chloropropane.

State the reagent(s) and conditions needed for this synthesis.

reagent(s) ....

conditions ....

[2]

(d) A mixture of alkenes is formed from a halogenoalkane by an elimination reaction.

Name the **three** alkenes that are formed from 2-iodobutane.

....

....

....

.... [2]

(e) Name the type of compound that is responsible for the depletion of the ozone layer in the atmosphere.

Explain how this type of compound causes ozone depletion.

....

....

....

....

....

.... [3]

4 (c)	(conc) H_2SO_4 **and** (solid) KCl or (conc) HCl heat /warm/T/ reflux	1 1 **[2]**	**ALLOW** PCl_3 / PCl_5 / $SOCl_2$ **NOT** (aq) **NOT** dil **ALLOW** (aq)
4 (d)	but-1-ene cis / Z- **and** trans / E-but-2-ene	1 1 **[2]**	
4 (e)	Chlorofluorocarbons / freons / CFCs	1	**ALLOW** NO

	<u>UV</u> causes the formation of chlorine free radicals / atoms / breaks C–Cl bonds (homolytically) Free radical reacts with O_3 to make O_2 (and •ClO)	1 1 **[3]**	**ALLOW** NO is a free radical **ALLOW** free radical reacts with O_3 to make O_2 (and NO_2) **ALLOW** free radical reacts with O_3

Candidate A answer

(c) Propan-1-ol is used to synthesise 1-chloropropane.

State the reagent(s) and conditions needed for this synthesis.

reagent(s) HCl ✓

conditions heat, temperature ✓

[2] 2

(d) A mixture of alkenes is formed from a halogenoalkane by an elimination reaction.

Name the **three** alkenes that are formed from 2-iodobutane.

For Examiner's Use

but-1-ene, cis-but-2-ene, trans-but-2-ene

✓ ✓ ✓

1

[2] 2

(e) Name the type of compound that is responsible for the depletion of the ozone layer in the atmosphere.

Explain how this type of compound causes ozone depletion.

Type : chloro fluoro ~~car~~ carbons (CFC), haloalkanes

at normal conditions O_3 is converted into O_2 and $O\cdot$

this reaction is reversible, but CFC's form chlorine radical

when ~~that~~ uv light is applied : $CF_2Cl_2 \xrightarrow{uv} CF_2Cl\cdot + Cl\cdot$

~~$Cl\cdot$~~ reactive chlorine radical reacts with ozone

$Cl\cdot + O_3 \rightarrow O_2 + ClO\cdot$ [3]

$ClO\cdot + O_3 \rightarrow 2O_2 + Cl\cdot$

Chlorine radical is regenerated, so it causes continious reaction which is not reversible

overalle reaction is

$2O_3 \longrightarrow 3O_2$

3

Comments for candidate A answer

4c

Candidate correct defined the reagent(s) and conditions needed for synthesis of 1-chloropropane from propan-1-ol

4d

Candidate correct named all of **three** alkenes that are formed from 2-iodobutane

4e

Candidate correct named the type of compound that is responsible for the depletion of the ozone layer in the atmosphere. The candidate completely explained how of this type compound causes ozone depletion.

Candidate C answer

(c) Propan-1-ol is used to synthesise 1-chloropropane.

State the reagent(s) and conditions needed for this synthesis.

reagent(s) PCl_5 ✓

conditions warm conditions ✗

[2] 1

(d) A mixture of alkenes is formed from a halogenoalkane by an elimination reaction.

For Examiner's Use

Name the **three** alkenes that are formed from 2-iodobutane.

but-1-ene, cis-but-2-ene, trans-but-2-ene

✓ ✓ ✓

1

[2] 2

(e) Name the type of compound that is responsible for the depletion of the ozone layer in the atmosphere.

Explain how this type of compound causes ozone depletion. ✓

✓ CFC - chlorofluorocarbons. ✓ Radicals react with oxone and produce active oxygen. This active oxygen is responsible for ozone depletion ✗

$Cl\cdot + O_3 \rightarrow ClO^- + O_2$

$ClO^- + O_2 \rightarrow ClO_2 + O$ [3] 2

Comments for candidate C answer

4c

Candidate correct defined the reagent(s) but could't gave correct conditions needed for synthesis of 1-chloropropane from propan-1-ol

4d

Candidate correct named all of **three** alkenes that are formed from 2-iodobutane

4e

Candidate correct named the type of compound that is responsible for the depletion of the ozone layer in the atmosphere. The candidate incompletely explain how of this type compound causes ozone depletion with relevant key words.

Candidate E answer

(c) Propan-1-ol is used to synthesise 1-chloropropane.

State the reagent(s) and conditions needed for this synthesis.

reagent(s) NaOH ~~(warm, OH⁻ as nucleophile)~~

conditions warm (OH⁻ act as nucleophile)

[2]

1 1

(d) A mixture of alkenes is formed from a halogenoalkane by an elimination reaction.

Name the **three** alkenes that are formed from 2-iodobutane.

1) butene-1

2) but-2ene

3) but-1,3-ene

[2]

For Examiner's Use

1 1

(e) Name the type of compound that is responsible for the depletion of the ozone layer in the atmosphere.

Explain how this type of compound causes ozone depletion.

CFC - chlorofluorocarbons. Radicals react with oxone and produce active oxygen. This active oxygen is responsible for ozone depletion

$Cl\cdot + O_3 \rightarrow ClO^- + O_2$

$ClO^- + O_2 \rightarrow ClO_2 + O$

[3]

2

Comments for candidate E answer

4c

Candidate could't define the reagent(s) but gave correct conditions needed for synthesis of 1-chloropropane from propan-1-ol

4d

Candidate correct named just one of **three** alkenes that are formed from 2-iodobutane therefore lost 1 point

4e

Candidate correct named the type of compound that is responsible for the depletion of the ozone layer in the atmosphere. The candidate incompletely explain how of this type compound causes ozone depletion with relevant key words.

5 a,b questions

5 In a closed system, carbon monoxide and oxygen are in equilibrium with carbon dioxide as shown.

$$2CO(g) + O_2(g) \rightleftharpoons 2CO_2(g)$$

(a) The table shows some thermochemical data for carbon monoxide, oxygen and carbon dioxide.

substance	$\Delta H_f^{\ominus}$ / kJ mol^{-1}	$S^{\ominus}$ /J K^{-1} mol^{-1}
$CO(g)$	–110	197
$O_2(g)$	0	205
$CO_2(g)$	–393	214

(i) Calculate the standard enthalpy change, $\Delta H_r^{\ominus}$, for this reaction.

$\Delta H_r^{\ominus}$ .... kJ mol^{-1} [2]

(ii) Calculate the standard entropy change, $\Delta S_r^{\ominus}$, for this reaction.

$\Delta S_r^\ominus$ .... $J\ K^{-1}\ mol^{-1}$ [2]

(b) The temperature of an equilibrium mixture of CO, O_2 and CO_2 is increased.

Explain what happens to the position of equilibrium.

Use your answer to **(a)(i)**.

....

....

.... [2]

5 (a) (i)	For correct equation or calculation $\Delta H_r = 2\Delta H_f CO_2 - 2\Delta H_f CO$ **or** $\Delta H_r = 2(-393) - 2(-110)$ $= -566\ (kJ\ mol^{-1})$	1 1 **[2]**	Answer = 2/2 **ALLOW** –570 –283 = 1/2 (+)566/570 = 1/2
5 (a) (ii)	For correct equation or calculation $\Delta S = 2SCO_2 - (SO_2 + 2SCO)$ **or** $\Delta S_r = 2(214) - 205 - 2(197)$ $= -171$	1 1 **[2]**	Answer = 2/2 **ALLOW** –170 –188 = 1/2 (+)171/170 = 1/2
5 (b)	(position of) equilibrium shifts to the left / shifts to the side of the reactants / shifts to carbon dioxide decomposition to oppose the effect of increasing temperature / because the reverse reaction	1 1	**ALLOW** ecf from endothermic in **(a)(i)** **ALLOW** (forward) reaction is exothermic /

	is endothermic / because the reverse reaction absorbs energy	**[2]**	releases energy

Candidate A answer

5 In a closed system, carbon monoxide and oxygen are in equilibrium with carbon dioxide as shown.

$$2CO(g) + O_2(g) \rightleftharpoons 2CO_2(g)$$

For Examiner's Use

(a) The table shows some thermochemical data for carbon monoxide, oxygen and carbon dioxide.

substance	**$\Delta H_f^{\ominus}$ / kJ mol^{-1}**	**$S^{\ominus}$ /J K^{-1} mol^{-1}**
CO(g)	–110	197
O_2(g)	0	205
CO_2(g)	–393	214

(i) Calculate the standard enthalpy change, $\Delta H_r^{\ominus}$, for this reaction.

$\Delta H_r^{\ominus} = \Delta H_{prod} - \Delta H_{react}$ ✓

$\Delta H_r^{\ominus} = 2 \times (-393) - (2 \times (-110) + 0) = -566$ kJmol^{-1}

$\Delta H_r^{\ominus}$ ......-566...... kJ mol^{-1} [2]

2

(ii) Calculate the standard entropy change, $\Delta S_r^{\ominus}$, for this reaction.

$\Delta S_r^{\ominus} = \Delta S_{prod} - \Delta S_{react}$

$\Delta S_r^{\ominus} = 2 \times 214 - (2 \times 197 + 205) = -171 \ J K^{-1} mol^{-1}$

$\Delta S_r^{\ominus}$ ......-171...... J K^{-1} mol^{-1} [2]

2

(b) The temperature of an equilibrium mixture of CO, O_2 and CO_2 is increased.

Explain what happens to the position of equilibrium.

Use your answer to **(a)(i)**.

Equillibrium shifts to the left because reaction is exothermic ($\Delta H_r = -566$ kJmol^{-1})

.... [2]

2

Comments for candidate A answer

5a(i)

Candidate used correct equation for calculation the standard enthalpy change, ∆Hr, for this reaction and gave answer with correct sign

5a(ii)

Candidate used correct equation for calculation the standard entropy change, $\Delta S_r^{\ominus}$, for this reaction and gave answer with correct sign

5b

Candidate used your answer to (a)(i) for completely explain the position of equilibrium with increasing temperature

Candidate C answer

5 In a closed system, carbon monoxide and oxygen are in equilibrium with carbon dioxide as shown.

$$2CO(g) + O_2(g) \rightleftharpoons 2CO_2(g)$$

(a) The table shows some thermochemical data for carbon monoxide, oxygen and carbon dioxide.

substance	$\Delta H_f^\ominus$ / kJ mol^{-1}	$S^\ominus$ /J K^{-1} mol^{-1}
CO(g)	−110	2 197
O_2(g)	0	205
CO_2(g)	2x −393	2 214

(i) Calculate the standard enthalpy change, $\Delta H_r^\ominus$, for this reaction.

$\Delta H_r^\ominus = 2 \times (-393) - (-220) = -566$ kJ mol^{-1}

$\Delta H_r^\ominus$ = −566 kJ mol^{-1} [2]

2 2

(ii) Calculate the standard entropy change, $\Delta S_r^\ominus$, for this reaction.

$\Delta S = (394 + 205) - 428 = 171$ J K^{-1} mol^{-1}

$\Delta S_r^\ominus$ = 171 J K^{-1} mol^{-1} [2]

1 1

(b) The temperature of an equilibrium mixture of CO, O_2 and CO_2 is increased.

Explain what happens to the position of equilibrium.

Use your answer to **(a)(i)**.

By the Le Chartelier principle by increasing temperature in exothermic reaction equilibrium goes to the left. [2]

2 2

Comments for candidate C answer

5a(i)

Candidate used correct equation for calculation the standard enthalpy change, ∆Hr, for this reaction and gave answer with correct sign

5a(ii)

Candidate used incorrect equation for calculation the standard entropy change, $\Delta S_r^\ominus$, therefore gave answer with incorrect sign, hence got 1 point

5b

Candidate used your answer to (a)(i) for completely explain the position of equilibrium with increasing temperature

Candidate E answer

5 In a closed system, carbon monoxide and oxygen are in equilibrium with carbon dioxide as shown.

For Examiner's Use

$$2CO(g) + O_2(g) \rightleftharpoons 2CO_2(g)$$

(a) The table shows some thermochemical data for carbon monoxide, oxygen and carbon dioxide.

substance	$\Delta H_f^\ominus$ / kJ mol^{-1}	$S^\ominus$ /J K^{-1} mol^{-1}
$CO(g)$	–110	197
$O_2(g)$	0	205
$CO_2(g)$	–393	214

(i) Calculate the standard enthalpy change, $\Delta H_r^\ominus$, for this reaction.

$\Delta H_f = 2 \cdot (-110) + 0 \; 2 \cdot (-393)$

$\Delta H_r^\ominus = -220 + 786$ ✓ ✓

$\Delta H_r^\ominus = 566 kJ\ mol^{-1}$

$\Delta H_r^{\ominus}$ ......566...... X X kJ mol^{-1} [2] 1

(ii) Calculate the standard entropy change, $\Delta S_r^{\ominus}$, for this reaction.

197·2 + 205 = 214·2

Sr = 599 = 428 ✓ ✓

Sr = 171

599 - 428

$\Delta S_r^{\ominus}$ ......171...... X X J K^{-1} mol^{-1} [2] 1

(b) The temperature of an equilibrium mixture of CO, O_2 and CO_2 is increased.

Explain what happens to the position of equilibrium.

Use your answer to **(a)(i)**.

....

....

.... X X [2] 0

X X

Comments for candidate E answer

5a(i)

Candidate used incorrect equation for calculation the standard enthalpy change, ΔHr, , therefore gave answer with incorrect sign, hence got 1 point

5a(ii)

Candidate used incorrect equation for calculation the standard entropy change, $\Delta S_r^{\ominus}$, therefore gave answer with incorrect sign, hence got 1 point

5b

Candidate doesn't explain the position of equilibrium with increasing temperature therefore got 0 mark

5c question

(c) Calculate the free energy change, ΔG, in kJ mol^{-1}, for the reaction between CO and O_2 at 100 °C.

Explain whether this reaction is feasible at 100 °C.

Use your answers to **(a)(i)** and **(a)(ii)**.

ΔG = ... kJ mol^{-1}

explanation ....

....

....

[3]

5 (c)	$\Delta G = \Delta H - T\Delta S$ **or** $\Delta G = -566 - (373 \times -171 \times 10^{-3})$ / $-566 - (373 \times -170 \times 10^{-3})$	1 1	 **ALLOW** -502 kJ mol^{-1} ecf from **a(i)** and **a(ii)**

	= –502.22 kJ mol^{-1}/–502.22 x 10^3 J mol^{-1} /-502.6 kJ mol^{-1} The reaction is feasible because ΔG <0 / –ve	1 **[3]**	–502220/502000 or –548.9 = 1 of M1 and M2 only **ALLOW** ecf from + sign in M2

Candidate A answer

(c) Calculate the free energy change, ΔG, in kJ mol^{-1}, for the reaction between CO and O_2 at 100 °C. *For Examiner's Use*

Explain whether this reaction is feasible at 100 °C.

Use your answers to **(a)(i)** and **(a)(ii)**.

$\Delta G = \Delta H - T \cdot \Delta S = -566000 + 373 \cdot 171 =$

$= -502217 J/mol = -502.2 kJ mol^{-1}$

$\Delta G =$ -502.2 kJ mol^{-1}

explanation $\Delta G < 0$ that is why reaction is feasible

~~on~~ because entropy increases

[3] 3 3

Comments for candidate A answer

5c

Candidate wrote correct expression and used your answers to (a)(i) and (a)(ii) for calculation free energy change, ΔG, in kJ mol□1, for the reaction. The candidate completely explaned feasibility of reaction at 100 □C based value free energy change

Candidate C answer

(c) Calculate the free energy change, ΔG, in kJ mol^{-1}, for the reaction between CO and O_2 at 100 °C. *For Examiner's Use*

Explain whether this reaction is feasible at 100 °C.

Use your answers to **(a)(i)** and **(a)(ii)**. $100°C = 373 K$

$\Delta G = \Delta H - T\Delta S = -566 - 373 \cdot 238 = -88713 kJ mol^{-1}$

explanation reaction is feasible because when ΔG is negative, reaction is spontaneous and there is high disorder

e.c.f ✓

2

Comments for candidate C answer

5c

Candidate wrote correct expression and used your answers to (a)(i) and (a)(ii) for calculation free energy change, ΔG, in kJ mol□1, for the reaction, but incorrect units entropy hence ecf candidate explained feasibility of reaction at 100 □C based value free energy change and got 2 point

Candidate E answer

11

(c) Calculate the free energy change, ΔG, in kJ mol^{-1}, for the reaction between CO and O_2 at 100 °C.

Explain whether this reaction is feasible at 100 °C.

Use your answers to **(a)(i)** and **(a)(ii)**.

For Examiner's Use

$\Delta G = \Delta H - T\Delta S$ ✓

$\Delta G = 566 \cancel{- 100 \cdot 1027 = -102700}$

$= 566 - 100 \cdot 1027 \cdot 10^3 =$

$= -102699434$

$\Delta S = 1027 \cdot 10^3 = 1027000$ ~~kJ mol~~ $kJ^{-1} mol^{-1}$

ΔG = -102699434 kJ mol^{-1} ✗

explanation the reaction is not feasible at 100°C, because the Gibbs free energy value is too negative ~~enough~~. ✗

[3] 1

Comments for candidate A answer

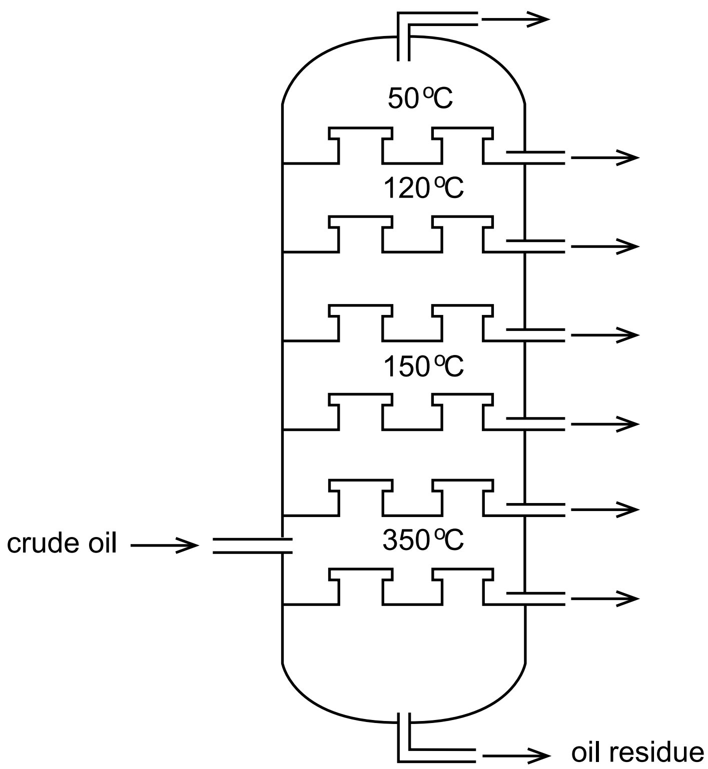
5c

Candidate wrote correct expression and used incorrect your answers to (a)(i) and (a)(ii) for calculation free energy change, ΔG, in kJ mol□1, for the reaction and gave incorrect explanation of feasibility of reaction

6a,b(i) questions

6 Kazakhstan has large oil and gas reserves.

(a) Crude oil undergoes fractional distillation to form different products. A fractionating column is shown.



Name the crude oil fraction that boils between 100 and 200 ºC.

State **one** use of this fraction.

name ....

use ....

[2]

(b) The petrochemical industry makes many important organic compounds.

(i) Cracking is one of the processes used to make these petrochemicals.

Name the two types of cracking.

....

.... [1]

6 (a)	gasoline / petrol **or** naphtha **or** kerosene / paraffin fuel for transport **or** a feedstock for petrochemicals **or** aircraft fuel / cooking stoves / heating stoves	1 1 **[2]**	use must match fraction name chosen **ALLOW** ecf from incorrect fraction name
6 (b) (i)	Thermal (cracking) **and** Catalytic (cracking)	1 **[1]**	Both needed for 1 mark

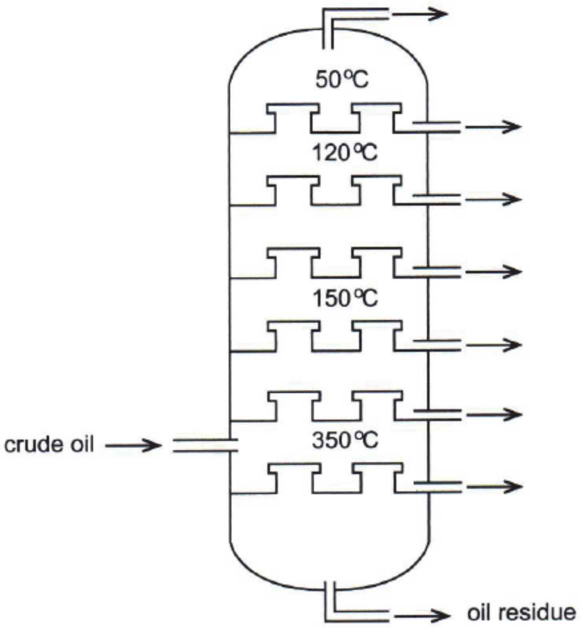
Candidate A answer

12

6 Kazakhstan has large oil and gas reserves.

(a) Crude oil undergoes fractional distillation to form different products. A fractionating column is shown.

For Examiner Use



name ~~gasoline~~ gasoline, petroleum

use fuel for cars

[2] 2

(b) The petrochemical industry makes many important organic compounds.

(i) Cracking is one of the processes used to make these petrochemicals.

Name the two types of cracking.

Catalytic cracking and thermal cracking. } 1

[1] 1

Comments for candidate A answer

6a

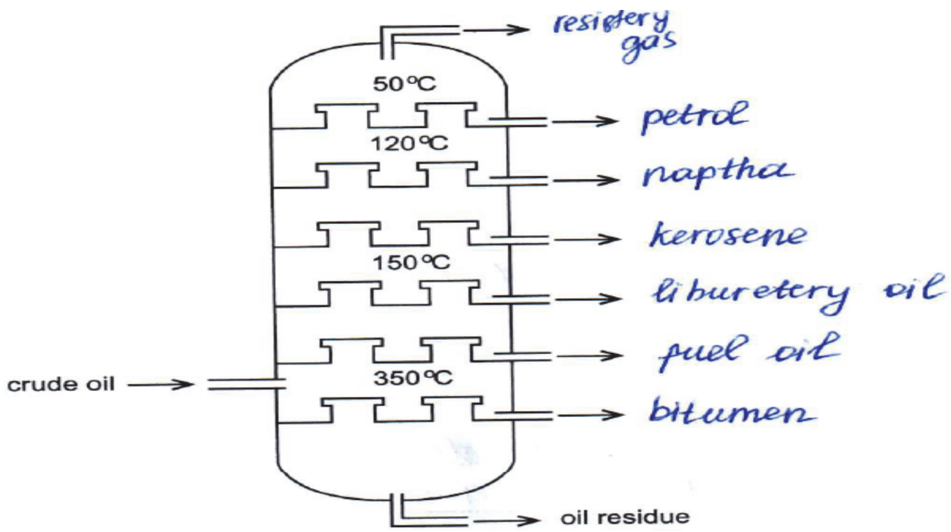
6b

Candidate C answer

6 Kazakhstan has large oil and gas reserves.

For Examiner's Use

(a) Crude oil undergoes fractional distillation to form different products. A fractionating column is shown.



Name the crude oil fraction that boils between 100 and 200 °C.

State **one** use of this fraction.

name ....kerosene ✓........

use ....kerosene lamps ✗........

[2] 1

(b) The petrochemical industry makes many important organic compounds.

(i) Cracking is one of the processes used to make these petrochemicals.

Name the two types of cracking.

........Catalytic ✓ cracking and thermal ✓ cracking. } 1........ [1] 1

Comments for candidate C answer

6a

6b

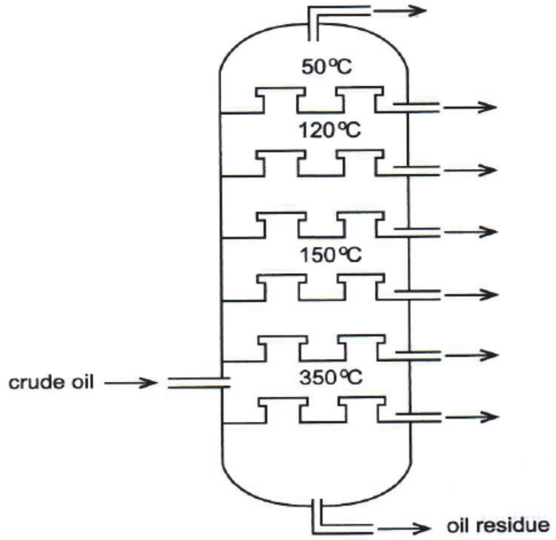
Candidate E answer

12

6 Kazakhstan has large oil and gas reserves.

For Examiner's Use

(a) Crude oil undergoes fractional distillation to form different products. A fractionating column is shown.



Name the crude oil fraction that boils between 100 and 200 °C.

State **one** use of this fraction.

name ....fuel for cars , kerosene....

use ....in ~~the~~ cars....

[2] 1 1

(b) The petrochemical industry makes many important organic compounds.

(i) Cracking is one of the processes used to make these petrochemicals.

Name the two types of cracking.

....Thermal and ~~te~~rmal....

.... [1] 0 0

Comments for candidate E answer

6a

6b

6b(ii,iii,iv) questions

6b(ii) Reforming is another process used to make petrochemicals.

During reforming the number of carbon atoms per hydrocarbon molecule stays the same but the number of hydrogen atoms is decreased.

Construct equations to show the reforming of C_6H_{14} to give:

hex-1-ene

....

....

benzene.

....

[2]

(iii) A hydrocarbon, $C_{21}H_{44}$, is cracked to form ethene, propene and one other hydrocarbon, **X**.

Ethene and propene are formed in the mole ratio 3:2 respectively.

Construct the balanced equation for this reaction.

.... [1]

(iv) Construct the balanced equation for the incomplete combustion of octane, C_8H_{18}.

.... [2]

6 (b) (ii)	$C_6H_{14} \rightarrow C_6H_{12} + H_2$ $C_6H_{14} \rightarrow C_6H_6 + 4H_2$	1 1 **[2]**	**ALLOW** any unambiguous formulae
6 (b) (iii)	$C_{21}H_{44} \rightarrow 3C_2H_4 + 2C_3H_6 + C_9H_{20}$	1 **[1]**	**IGNORE** state symbols
6 (b) (iv)	$C_8H_{18} + 4\frac{1}{2}O_2 \rightarrow 8C + 9H_2O$ **or** $C_8H_{18} + 8\frac{1}{2}O_2 \rightarrow 8CO + 9H_2O$ M1 = species M2 = balancing	1+1 **[2]**	**ALLOW** multiples **ALLOW** equations with both C and CO as products

Candidate A answer

(ii) Reforming is another process used to make petrochemicals.

During reforming the number of carbon atoms per hydrocarbon molecule stays the same but the number of hydrogen atoms is decreased.

Construct equations to show the reforming of C_6H_{14} to give:

hex-1-ene $C_6H_{14} \xrightarrow{\text{reforming}} C_6H_{12} + H_2$

benzene. $C_6H_{14} \xrightarrow{\text{ref}}$ ~~$C_6H_{12} + H_2$~~

$C_6H_6 + 4H_2$ [2]

For Examiner's Use

2

(iii) A hydrocarbon, $C_{21}H_{44}$, is cracked to form ethene, propene and one other hydrocarbon, **X**.

Ethene and propene are formed in the mole ratio 3:2 respectively.

Construct the balanced equation for this reaction.

$C_{21}H_{44} \rightarrow 3C_2H_4 + 2C_3H_6 + C_9H_{20}$ [1]

1

(iv) Construct the balanced equation for the incomplete combustion of octane, C_8H_{18}.

$C_8H_{18(l)} + \frac{17}{2}O_{2(g)} \rightarrow 8CO_{(g)} + 9H_2O_{(l)}$ [2]

2

Comments for candidate A answer

6b(ii)

6b(iii)

6b(iv)

Candidate C answer

(ii) Reforming is another process used to make petrochemicals.

During reforming the number of carbon atoms per hydrocarbon molecule stays the same but the number of hydrogen atoms is decreased.

Construct equations to show the reforming of C_6H_{14} to give:

hex-1-ene

$C_6H_{14} \rightarrow C_6H_{12} + H_2$ ✓

benzene. C_6H_6

$C_6H_{14} \rightarrow C_6H_6 + H_8$ ✗

[2] 1

(iii) A hydrocarbon, $C_{21}H_{44}$, is cracked to form ethene, propene and one other hydrocarbon, **X**.

Ethene and propene are formed in the mole ratio 3:2 respectively.

Construct the balanced equation for this reaction.

$C_{21}H_{44} \rightarrow 3C_2H_4 + 2C_3H_6 + C_9H_{20}$ ✓ [1] 1

(iv) Construct the balanced equation for the incomplete combustion of octane, C_8H_{18}.

✓ $C_8H_{18} + \frac{17}{2}O_2 \rightarrow 8CO + 9H_2O$ ✓ [2] 2

Comments for candidate C answer

6b(ii)

6b(iii)

6b(iv)

Candidate E answer

$-C=C-C-C-C-C$ $H-C=C-C-H$

(ii) Reforming is another process used to make petrochemicals.

During reforming the number of carbon atoms per hydrocarbon molecule stays the same but the number of hydrogen atoms is decreased.

Construct equations to show the reforming of C_6H_{14} to give:

hex-1-ene

$C_6H_{14} - H_2 \rightarrow C_6H_{12}$ ✗

benzene.

$C_6H_{14} - 4H_2 \rightarrow C_6H_6$ ✗

[2] 0

(iii) A hydrocarbon, $C_{21}H_{44}$, is cracked to form ethene, propene and one other hydrocarbon, **X**.

Ethene and propene are formed in the mole ratio 3:2 respectively.

Construct the balanced equation for this reaction.

$C_{21}H_{44} \xrightarrow{t°c} 3C_2H_4 + 2C_3H_6 + C_9H_{20}$ ✓ [1] 1

(iv) Construct the balanced equation for the incomplete combustion of octane, C_8H_{18}.

$C_8H_{18} + 8\frac{1}{2}O_2 \rightarrow 8CO + 9H_2O$ ✓ ✓ [2] 2

Comments for candidate E answer

6b(ii)

6b(iii)

6b(iv)

6c question

6(c) An ester has the structure:

CH$_2$OCOR
|
CHOCOR
|
CH$_2$OCOR

where R represents $–C_{15}H_{31}$.

The ester is boiled with aqueous sodium hydroxide to form a soap.

(i) Draw the structures of the two products.

[2]

(ii) Give the name of this type of reaction

.... [1]

6 (c) (i)	$CH_2OHCHOHCH_2OH$ $RCOO^-$ / $C_{15}H_{31}COO^-$ (Na^+)	1 1 **[2]**	**NOT** RCOOH / $C_{15}H_{31}COOH$
6 (c) (ii)	hydrolysis / saponification	1 **[1]**	

Candidate A answer

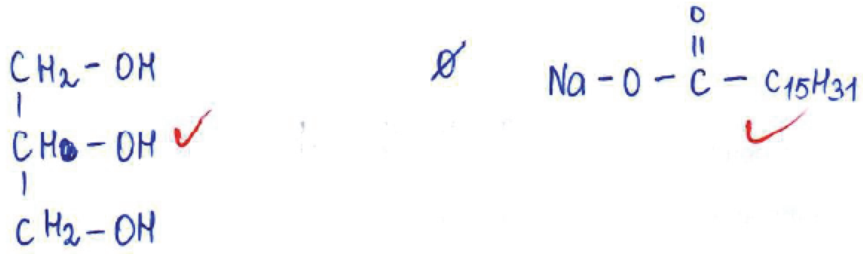
(c) An ester has the structure:

$$\begin{array}{l} CH_2OCOR \\ | \\ CHOCOR \\ | \\ CH_2OCOR \end{array}$$

where R represents $-C_{15}H_{31}$.

The ester is boiled with aqueous sodium hydroxide to form a soap.

(i) Draw the structures of the two products.



[2] 2

(ii) Give the name of this type of reaction

Basic hydrolysis ✓ [1] 1

[Total: 11]

For Examiner's Use

Comments for candidate A answer

6c(i)

6c(ii)

Candidate C answer

For Examiner Use

(c) An ester has the structure:

$$\begin{array}{l} CH_2OCOR \\ | \\ CHOCOR \\ | \\ CH_2OCOR \end{array} + 3NaOH \rightarrow$$

where R represents $-C_{15}H_{31}$.

The ester is boiled with aqueous sodium hydroxide to form a soap.

(i) Draw the structures of the two products.

$$\begin{array}{l} CH_2-OH \\ | \\ CH-OH \\ | \\ CH_2-OH \end{array} \checkmark \qquad C_{15}H_{31}C\begin{array}{l} {}^{\nearrow O} \\ {}_{\searrow O-Na} \end{array} \checkmark$$

[2] 2

(ii) Give the name of this type of reaction

.... ✗ [1] 0

Comments for candidate C answer

6c(i)

6c(ii)

Candidate E answer

For Examiner's Use

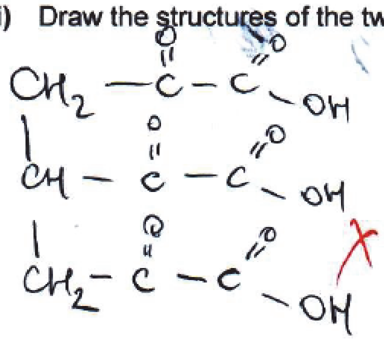
(c) An ester has the structure:

$$\begin{array}{l} CH_2OCOR \\ | \\ CHOCOR \\ | \\ CH_2OCOR \end{array}$$

where R represents $-C_{15}H_{31}$.

The ester is boiled with aqueous sodium hydroxide to form a soap.

(i) Draw the structures of the two products.



+ ~~3 NaR~~ 3 $C_{15}H_{31}^{-}Na^{+}$ ✗

[2] 0

(ii) Give the name of this type of reaction

(alkaline) hydrolysis ✓ [1] 1

[Total: 11]

Comments for candidate E answer

6c(i)

6c(ii)

7a,b,c,d questions

7 Amino acids contain at least two different functional groups.

In nature, most amino acids are 2-amino acids, otherwise known as α-amino acids.

(a) Draw the structural formula of 2-aminopropanoic acid.

[1]

(b) A few drops of universal indicator were added to aqueous potassium hydroxide. The mixture had a purple colour.

An aqueous solution of 2-amino-propanoic acid was slowly added to the purple mixture until it turned blue-green.

State **and** explain why the purple mixture turned blue-green.

....

....

....

.... [2]

(c) All α-amino acids except glycine are optically active.

Define the term *optically active*.

....

....

....

.... [1]

(d) The primary structure of a protein is held together by peptide links.

Draw the structure of a peptide link.

Show all of the atoms and all of the bonds.

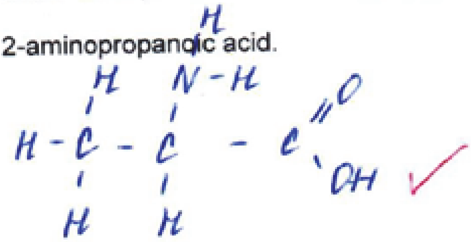
[1]

7 (a)	CH_3 – CH – COOH | NH_2	1 **[1]**	**ALLOW** displayed formula or a mix of displayed and structural formulae
7 (b)	KOH is neutralised / pH of the solution decreases acid functional group / –COOH group in the molecule **or** $CH_3CH(NH_2)COOH + OH^- \rightarrow CH_3CH(NH_2)COO^- + H_2O$	1 1 **[2]**	**ALLOW** equation for 2 marks
7 (c)	Rotates a beam of plane-polarised light / the plane of polarisation	1 **[1]**	
7 (d)	–C–N– || | O H	1 **[1]**	

Candidate A answer

7 Amino acids contain at least two different functional groups.
In nature, most amino acids are 2-amino acids, otherwise known as α-amino acids.

(a) Draw the structural formula of 2-aminopropanoic acid.



[1]

(b) A few drops of universal indicator were added to aqueous potassium hydroxide. The mixture had a purple colour.

An aqueous solution of 2-amino-propanoic acid was slowly added to the purple mixture until it turned blue-green.

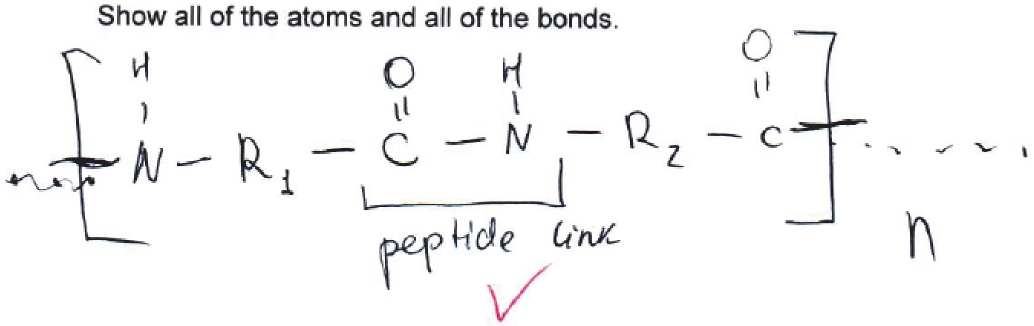
State **and** explain why the purple mixture turned blue-green.

amino acids exist as zwitter ions, in alkaline environment they react with OH^- ions, so conc. of OH^- ions decrease, it means that pH is decreased, amino acid acts as an acid, so neutralisation take place. [2]

(d) The primary structure of a protein is held together by peptide links.

Draw the structure of a peptide link.

Show all of the atoms and all of the bonds.



[1]

Comments for candidate A answer

7a

7b

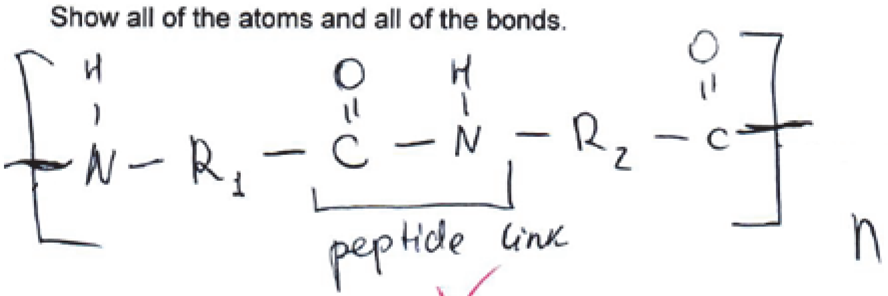
7d

Candidate C answer

(d) The primary structure of a protein is held together by peptide links.

Draw the structure of a peptide link.

Show all of the atoms and all of the bonds.

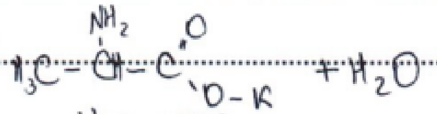


[1]

(b) A few drops of universal indicator were added to aqueous potassium hydroxide. The mixture had a purple colour.

An aqueous solution of 2-amino-propanoic acid was slowly added to the purple mixture until it turned blue-green.

State **and** explain why the purple mixture turned blue-green.



blue-green

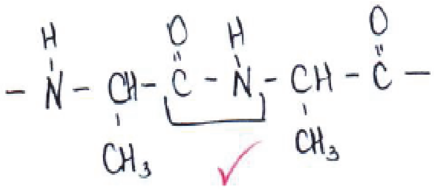
This is amphoteric, reacts with acid and base

[2]

(d) The primary structure of a protein is held together by peptide links.

Draw the structure of a peptide link.

Show all of the atoms and all of the bonds.



[1] 1

Comments for candidate C answer

7a

7b

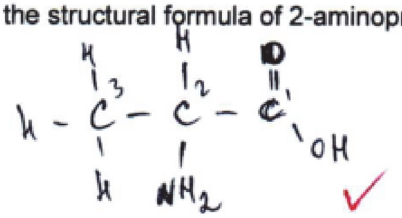
7d

Candidate E answer

7 Amino acids contain at least two different functional groups.
In nature, most amino acids are 2-amino acids, otherwise known as α-amino acids.

For Examiner's Use

(a) Draw the structural formula of 2-aminopropanoic acid.



(alanine)

[1] P

(b) A few drops of universal indicator were added to aqueous potassium hydroxide. The mixture had a purple colour.

An aqueous solution of 2-amino-propanoic acid was slowly added to the purple mixture until it turned blue-green.

State **and** explain why the purple mixture turned blue-green.

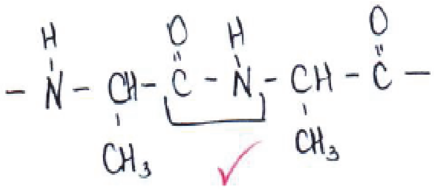
$KOH + CH_3CHNH_2COOH$

The ammonia will be produced causing the basic solution with an acid (2-aminopropanoic acid) to change the pH concentration and the indicator changes from a cold (purple) colour to a bright (blue-green) colour ✗ ✗ [2] 0

(d) The primary structure of a protein is held together by peptide links.

Draw the structure of a peptide link.

Show all of the atoms and all of the bonds.



[1]

Comments for candidate E answer

7a

7b

7d

7e,f questions

7 (e) Describe the meaning of the term *secondary structure of a protein*.

State the type of bonding involved in maintaining the secondary structure.

....

....

.... [2]

(f) Sickle cell anaemia is an example of a disease caused by a genetic disorder.

Describe the genetic basis of diseases.

....

....

....

....

....

....

....

.... [4]

7 (e)	(regular) folding of polypeptide chain hydrogen bonds	1 1 **[2]**	**ALLOW** reference to α-helix or β-pleated sheet
7 (f)	**Any 4 from** mutation changed base sequence in DNA one base changed for another / one base missing change of protein (primary) structure / destroy sequence of amino acids protein shape / function changed	 1 1 1 1 1 **[max 4]**	

Candidate A answer

(e) Describe the meaning of the term *secondary structure of a protein*.

State the type of bonding involved in maintaining the secondary structure.

For Examiner's Use

Secondary structure of protein - has hydrogen bond and have α-helix and β-sheets peptide bond [2]

2 2

(f) Sickle cell anaemia is an example of a disease caused by a genetic disorder.

Describe the genetic basis of diseases.

Sickle cell anaemia is a result of DNA mutation. In translation amino acid was improperly coded that is one nitrogenous base was substituted by another. resulted in wrong amino acid that leads to sickle cell anaemia. Such kind of mutations is called substitution. There are other kind of mutations like silent mutations when [4] nitrogenous base was coded improperly but it results in the same amino acid, so ~~it is not~~ the protein can function Non-sense mutation is when nitrogenous base was coded improperly, but it results in stop-codon, so translation is stopped. Also, deletion, when a nitrogenous base is deleted or insertion when nitrogenous base is added that results in genetic disorder

4

Comments for candidate E answer

7e

7f

Candidate C answer

(e) Describe the meaning of the term *secondary structure of a protein*.

State the type of bonding involved in maintaining the secondary structure.

Maintained by hydrogen bond ~~on~~ between

CO and NH of another peptide bond

[2]

For Examiner's Use

1

(f) Sickle cell anaemia is an example of a disease caused by a genetic disorder.

Describe the genetic basis of diseases.

Mutation

Changing of amino acids

Deletion ~~of~~ or substitution of nitrogen bases

Mistakes during the transcription;

The information will not be copied from DNA to mRNA regularly

[4]

3

Comments for candidate C answer

7e

7f

Candidate E answer

(e) Describe the meaning of the term *secondary structure of a protein*.

State the type of bonding involved in maintaining the secondary structure.

For Examiner's Use

Secondary structure is when the structure contains two peptide links ; the are bonded in peptide type and they have α-helix and β pleated sheets. [2]

1

(f) Sickle cell anaemia is an example of a disease caused by a genetic disorder.

Describe the genetic basis of diseases.

Sickle cell anaemia is caused by a genetic disorder. The genetic disorder happens when 3 successive (triplet) codons are mutated in the sequence. The insertion, deletion or substitution can happen at this moment and 1 of the triplet codons can change their positions or even leave it which will cause a disorder and confusion in producing the next chain. When [4] the stop-codon appears it stops functioning which means their will be a lack of protein and a big illness.

2

Comments for candidate E answer

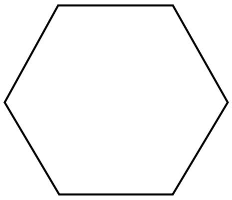
7e

7f

8a,b,c questions

8 Benzene is an aromatic compound.

(a) Complete the diagram of a benzene molecule. Draw the bonding present by showing the overlap of atomic orbitals that may be present. Label the types of bonds clearly.



[2]

(b) Nitration is an important process in organic synthesis.

(i) Give the full name of the mechanism for the nitration of benzene.

.... [1]

(ii) Show the mechanism for the nitration of benzene.

Include any relevant curly arrows and charges.

[5]

(c) Nitrobenzene is reduced by heating with tin and concentrated hydrochloric acid.

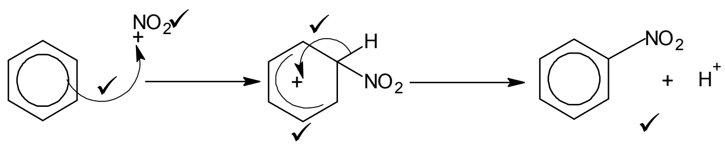
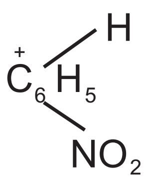
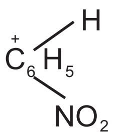
Write the balanced equation for the reduction of nitrobenzene.

Name the organic product formed.

equation ....

name ....

[2]

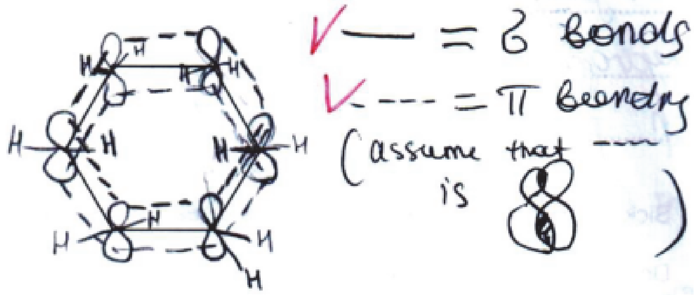
8 (a)	sigma bond(s) labelled overlapping p orbitals or 'clouds' above and below ring labelled as pi bond(ing) / delocalised pi-electron system	1 1 **[2]**	**IGNORE** hydrogen atoms
8 (b) (i)	electrophilic substitution	1 **[1]**	Both terms needed
8 (b) (ii)	 **OR** $2H_2SO_4 + HNO_3 \leftrightarrow H_30^+ + 2HSO_4^- + NO_2^+$ electrophilic attack $C_6H_6 + NO_2^+ \rightarrow$  elimination of proton  $+HSO_4^- \rightarrow C_6H_5NO_2 + H_2SO_4$	1 1 1 1 1 **[5]**	M1 curly arrow from ring to N M2 +NO2/NO2+ M3 intermediate cation M4 curly arrow from C–H into ring M5 both products
8 (c)	$C_6H_5NO_2 + 6[H] \rightarrow C_6H_5NH_2 + 2H_2O$	1	**ALLOW** phenylammonium

	aminobenzene / phenylamine	1 **[2]**	chloride/ aniline

Candidate A answer

8 Benzene is an aromatic compound.

(a) Complete the diagram of a benzene molecule. Draw the bonding present by showing the overlap of atomic orbitals that may be present. Label the types of bonds clearly.



[2] 2

For Examiner's Use

(b) Nitration is an important process in organic synthesis.

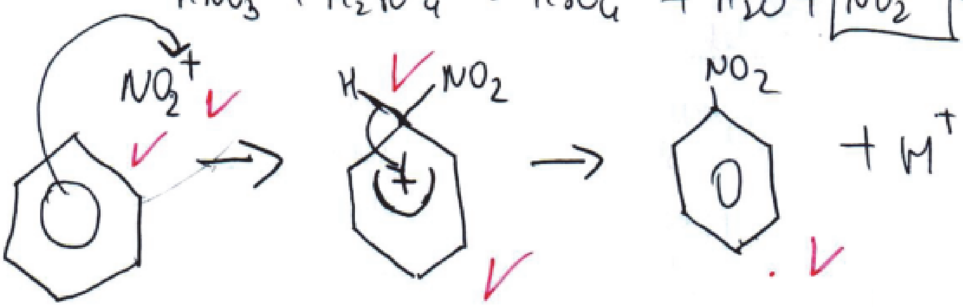
(i) Give the full name of the mechanism for the nitration of benzene.

electrophilic substitution ✓ [1] 1

(ii) Show the mechanism for the nitration of benzene.

Include any relevant curly arrows and charges.

$HNO_3 + H_2SO_4 \rightarrow HSO_4^- + H_2O + \boxed{NO_2^+}$ → electrophile



$H^+ + HSO_4^- \rightarrow H_2SO_4$ regeneration of catalyst

[5] 5

(c) Nitrobenzene is reduced by heating with tin and concentrated hydrochloric acid.

Write the balanced equation for the reduction of nitrobenzene.

Name the organic product formed.

equation $C_6H_5NO_2 + 6[H] \longrightarrow C_6H_5NH_2 + 2H_2O$ ✓

name aniline / aminobenzene ✓

[2] 9

Comments for candidate A answer

8a

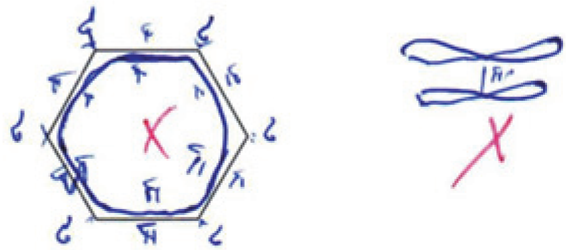
8b

8c

Candidate C answer

8 Benzene is an aromatic compound.

(a) Complete the diagram of a benzene molecule. Draw the bonding present by showing the overlap of atomic orbitals that may be present. Label the types of bonds clearly.



[2]

For Examiner Use

0

(b) Nitration is an important process in organic synthesis.

(i) Give the full name of the mechanism for the nitration of benzene.

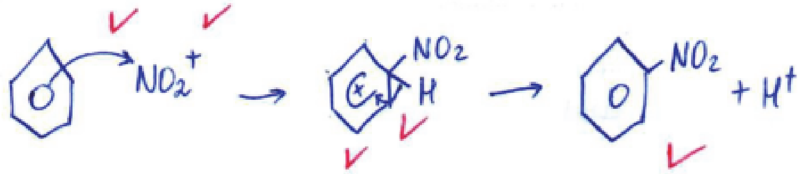
electrophillic substitution [1] 1

(ii) Show the mechanism for the nitration of benzene.

Include any relevant curly arrows and charges.

1) $HNO_{3\,conc} + H_2SO_{4\,conc} \rightarrow HSO_4^- + NO_2^+ + H_2O$

2)



3) $HSO_4^- + H^+ \rightarrow H_2SO_4$

[5] 5

(c) Nitrobenzene is reduced by heating with tin and concentrated hydrochloric acid.

Write the balanced equation for the reduction of nitrobenzene.

Name the organic product formed.

equation $C_6H_5NO_2 + 6[H] \longrightarrow C_6H_5NH_2 + 2H_2O$

name aniline / aminobenzene

[2] 9

Comments for candidate C answer

8a

8b

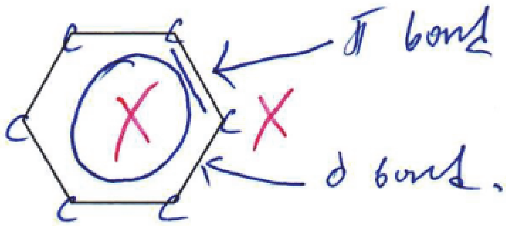
8c

Candidate E answer

8 Benzene is an aromatic compound.

For Examiner's Use

(a) Complete the diagram of a benzene molecule. Draw the bonding present by showing the overlap of atomic orbitals that may be present. Label the types of bonds clearly.



[2] 0

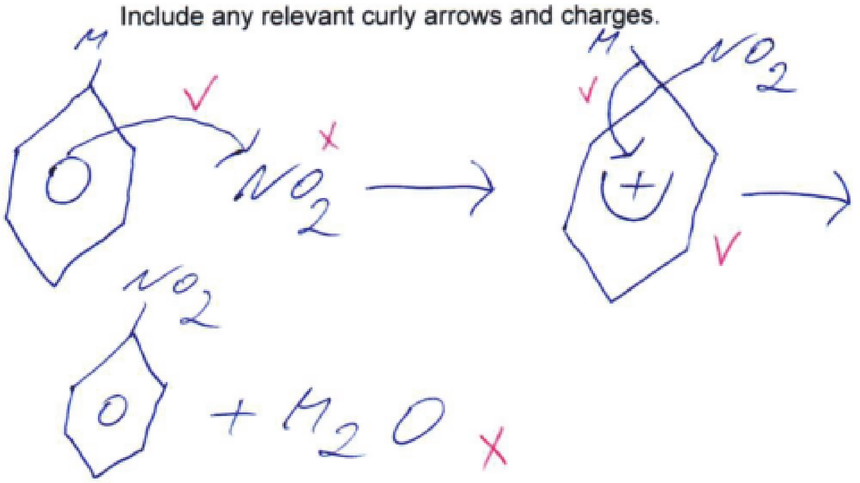
(b) Nitration is an important process in organic synthesis.

(i) Give the full name of the mechanism for the nitration of benzene.

electrophilic substitution [1] 1

(ii) Show the mechanism for the nitration of benzene.

Include any relevant curly arrows and charges.



[5] 3

(c) Nitrobenzene is reduced by heating with tin and concentrated hydrochloric acid.

Write the balanced equation for the reduction of nitrobenzene.

Name the organic product formed.

equation $C_6H_5NH_2 + HCl \xrightarrow{t^\circ\ Sn} C_6H_5Cl + NH_3$

name 1- Chlorobenzene

[2] 0

Comments for candidate E answer

8a

8b

8c

9a,b,c questions

9 Analytical techniques are very important in modern chemistry.

(a) Describe one use of gas-liquid chromatography in analytical chemistry.

.... [1]

(b) Describe how gas-liquid chromatography is used to separate organic compounds.

....

....

....

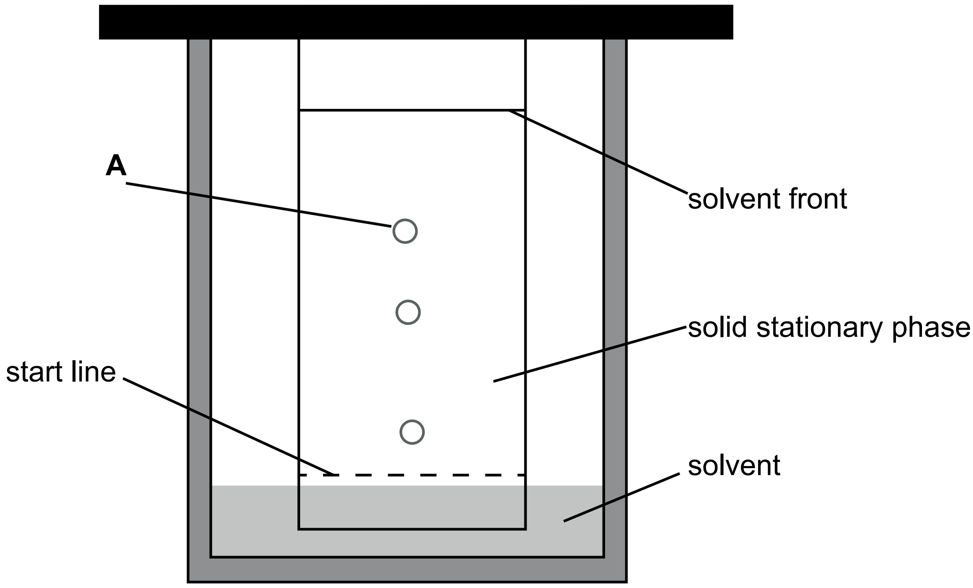
....

....

.... [3]

(c) Thin-layer chromatography is similar to paper chromatography.

The diagram shows the result of thin-layer chromatography of a mixture of metal ions.



(i) Describe how to calculate the R_f value of **A**.

....

....

.... [1]

(ii) The mixture of metal ions was spotted onto the start line.

Explain why the start line must be above the level of the solvent.

....

.... [1]

9 (a)	detecting banned substances in sport / food / medicine	1 **[1]**	**NOT** chemical analysis **ALLOW** any relevant answer
9 (b)	the vaporised sample is injected into the column by the carrier gas the stationary phase is a liquid coating absorbed on an inert solid	1 1 1	**ALLOW** owtte

	different components of the mixture have different retention times	**[3]**	
9 (c) (i)	Rf = $\dfrac{\text{distance moved by A}}{\text{distance moved by solvent front}}$	1 **[1]**	**ALLOW** in words **ALLOW** as labels on diagram
9 (c) (ii)	to avoid the sample / spot washing away into the solvent	1 **[1]**	**ALLOW owtte**

Candidate A answer

9 Analytical techniques are very important in modern chemistry.

(a) Describe one use of gas-liquid chromatography in analytical chemistry.

(to analyse the urine of sportsmen ~~when~~ whether they use some drugs or not [1]
analyse the alcohol (present or absence) – mostly policemen use to identify drunk

For Examiner's Use

1 1

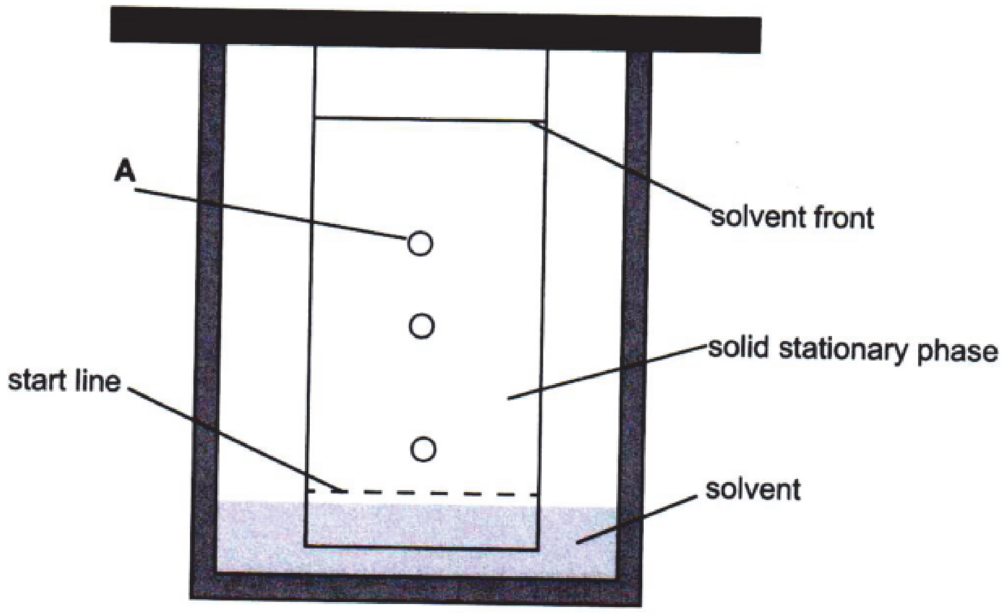
(b) Describe how gas-liquid chromatography is used to separate organic compounds.

There are two phases:
Stationary phase: ~~Silica Glass~~ Liquid alkanes
Mobile phase: ~~liquid alkanes~~ Inert gas (He)
~~This~~ If stationary phase is non-polar, polar compounds come out first. If stationary phase is polar, non-polar compounds come out first. Large, heavy particles have longer retention time than small, light particles. Organic substances are separated according to charge and mass [3]

2

(c) Thin-layer chromatography is similar to paper chromatography.

The diagram shows the result of thin-layer chromatography of a mixture of metal ions.



(i) Describe how to calculate the R_f value of **A**.

1) Calculate solute distance, solvent distance

2) Find ratio: $\frac{\text{solute distance}}{\text{solvent distance}}$ ✓

[1] 1

Solute distance - distance from start line to the centre of point A

Solvent distance - distace from start line to the line where solvent has reached on stationary phase

(ii) The mixture of metal ions was spotted onto the start line.

Explain why the start line must be above the level of the solvent.

Metal ions will dissolve ✓ in solvent before they are absorbed to paper /stationary phase [1]

For Examiner's Use

1

Comments for candidate A answer

9a

9b

9c

Candidate C answer

9 Analytical techniques are very important in modern chemistry.

For Examiner's Use

(a) Describe one use of gas-liquid chromatography in analytical chemistry.

criminalistics ✗ [1] 0

(b) Describe how gas-liquid chromatography is used to separate organic compounds.

There are two phases:

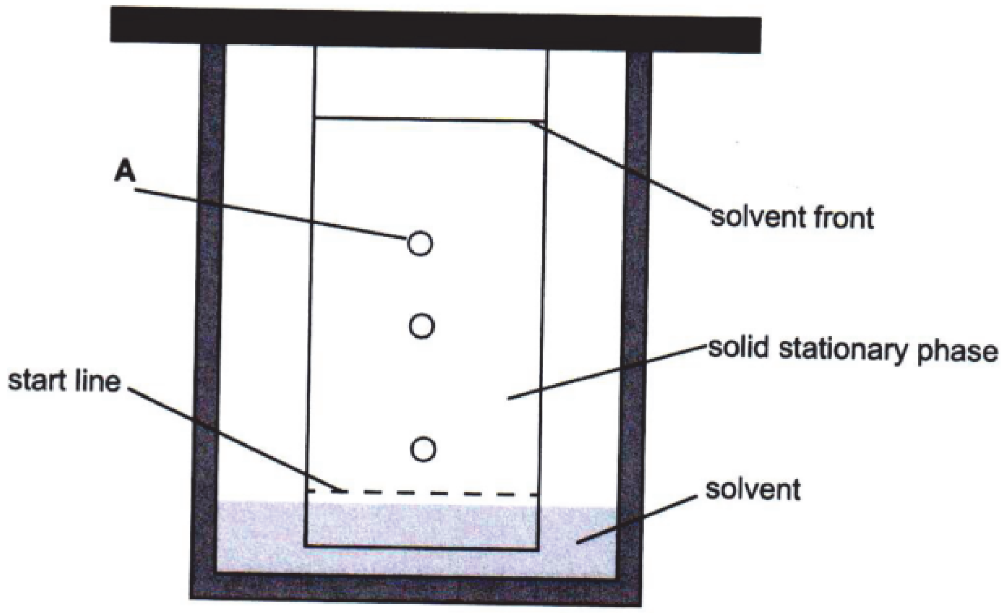
Stationary phase : ~~Solid Glass~~ Liquid alKanes ✗

Mobile phase ; ~~liquid carbons~~ Inert gas (He) ✓

~~This~~ If stationary phase is non-polar, polar compounds come out first. If stationary phase is polar, non-polar compounds come out first. Large, heavy particles have longer retention time ✓ than small, light particles. Organic substances are seperated according to charge and mass [3] 2

(c) Thin-layer chromatography is similar to paper chromatography.

The diagram shows the result of thin-layer chromatography of a mixture of metal ions.



(i) Describe how to calculate the R_f value of **A**.

$R_f = \dfrac{\text{distance reached by A}}{\text{distance by solvent}}$ ✓

[1] 1

Comments for candidate C answer

9a

9b

9c

Candidate E answer

9 Analytical techniques are very important in modern chemistry.

For Examiner's Use

(a) Describe one use of gas-liquid chromatography in analytical chemistry.

criminalistics X [1] 0

(b) Describe how gas-liquid chromatography is used to separate organic compounds.

There are two phases:

Stationary phase : ~~solid phase~~ Liquid alkanes ✗

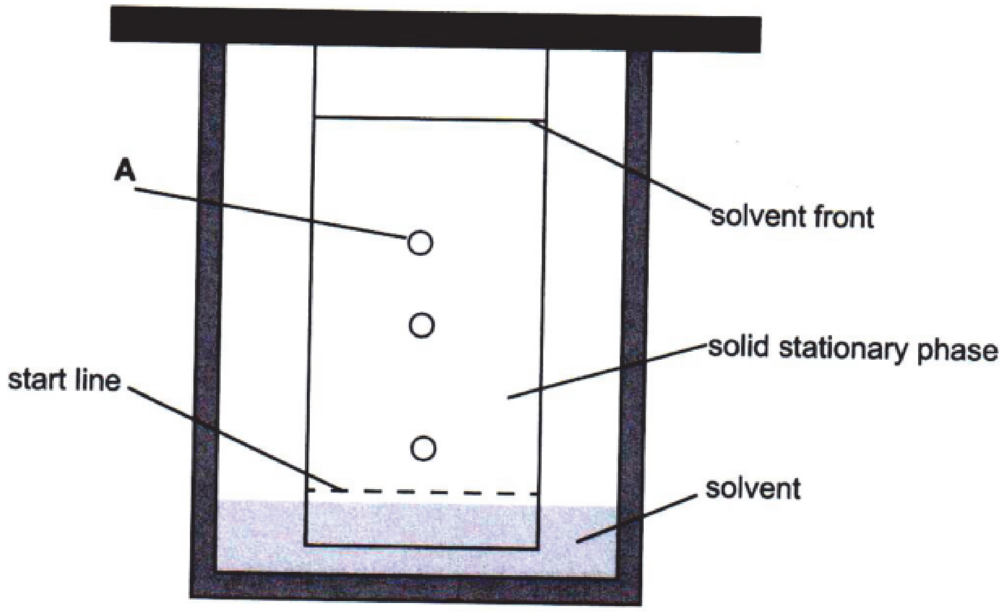
Mobile phase ; ~~liquid alkanes~~ Inert gas (He) ✓

~~This~~ If stationary phase is non-polar, polar compounds come out first. If stationary phase is polar, non-polar compounds come out first. Large, heavy particles have longer retention time ✓ than small, light particles. Organic substances are seperated according to charge and mass **[3]**

2

(c) Thin-layer chromatography is similar to paper chromatography.

The diagram shows the result of thin-layer chromatography of a mixture of metal ions.



(i) Describe how to calculate the R_f value of **A**.

measure the distance between start line and solvent front, and the distance between start line and point A and then divide **[1]**

$$R_f = \frac{\text{distance between start line and A}}{\text{distance between start line and solvent front}}$$ ✓

1

(ii) The mixture of metal ions was spotted onto the start line.

Explain why the start line must be above the level of the solvent.

~~When it is~~ to allow for mixture of unknown substances to go up, do not mix the mixture and solvent. [1]

For Examiner's Use

1

Comments for candidate E answer

9a

9b

9c

10a,b,c,d questions

10 A buffer solution is prepared by mixing a weak acid with one of its salts.

(a) Describe what is meant by the term *buffer solution*.

....

.... [1]

(b) The buffer solution contains 0.100 mol dm^{-3} HA and 0.150 mol dm^{-3} NaA.

The acid dissociation constant, K_a, for HA is 1.85×10^{-5} mol dm^{-3}.

Calculate the pH of this buffer solution.

pH = ... [3]

(c) Blood acts as a buffer solution.

Blood is a mixture that contains iron.

State the function of iron in blood.

....

.... [1]

(d) Blood also contains some proteins.

State **one** effect of toxic metals on proteins.

.... [1]

[Total: 6]

10 (a)	solution which resists changes in pH / pH hardly changes when small amounts of acid or alkali are added	1 **[1]**	**ALLOW** H^+ for acid **ALLOW** OH^- / base for alkali **NOT** maintains a constant pH
10 (b)	$1.85 \times 10^{-5} = \frac{[H^+] \times 0.150}{0.100}$ $[H+] = \frac{1.85 \times 10^{-5} \times 0.100}{0.150} = 1.23 \times 10^{-5}$ (mol dm^{-3}) pH = 4.91	1 1 1 **[3]**	1.23×10^{-5} = 2 marks 4.91 = 3 marks **ALLOW** ecf from incorrect $[H^+]$

10 (c)	to facilitate transport of oxygen in blood/in haemoglobin	1 **[1]**	
10 (d)	protein will change shape / protein will be denatured/change in properties	1 **[1]**	**ALLOW** destroy function of protein

Candidate A answer

10 A buffer solution is prepared by mixing a weak acid with one of its salts.

For Examiner's Use

(a) Describe what is meant by the term *buffer solution*.

Solution which resists pH changes when small amount of alkali or acid is added ✓ [1] 1

(b) The buffer solution contains 0.100 mol dm^{-3} HA and 0.150 mol dm^{-3} NaA.

The acid dissociation constant, K_a, for HA is 1.85 x 10^{-5} mol dm^{-3}.

Calculate the pH of this buffer solution.

$HA \rightleftarrows H^+ + A^-$

✓ $K_a = \frac{[H^+][A^-]}{[HA]} \Rightarrow [H^+] = \frac{K_a \cdot [HA]}{[A^-]}$

$[H^+] = \frac{1.85 \cdot 10^{-5} \cdot 0.1}{0.150} = 1.23 \cdot 10^{-5}$ ✓ $pH = -\log[H^+]$

pH = 4.91 ✓ [3] 3

(c) Blood acts as a buffer solution.

Blood is a mixture that contains iron.

State the function of iron in blood.

Oxygen transport by forming oxyhaemoglobin. [1] 1

(d) Blood also contains some proteins.

State **one** effect of toxic metals on proteins.

altering 3D shape and denaturation [1] 1
inhibition of enzymes.

Comments for candidate A answer

10a

10b

10c

10d

Candidate C answer

10 A buffer solution is prepared by mixing a weak acid with one of its salts.

For Examiner's Use

(a) Describe what is meant by the term *buffer solution*.

Buffer solution – solution that can minimize the effect of pH changes. [1] 0

(b) The buffer solution contains 0.100 mol dm^{-3} HA and 0.150 mol dm^{-3} NaA.

The acid dissociation constant, K_a, for HA is 1.85 x 10^{-5} mol dm^{-3}.

Calculate the pH of this buffer solution.

$[H^+] = K_a \cdot \frac{acid}{salt}$ $H^+ = 1{,}85 \cdot 10^{-5} \cdot \frac{0{,}1}{0{,}15} =$

$HCOOH + Na \rightarrow HCOONa + HCOO^-$ $1{,}23 \cdot 10^{-5}$

$c(HCOOH) = 0{,}1$

$pH = -\log(1{,}23 \cdot 10^{-5}) =$ 5

pH = 5 [3] 2

(c) Blood acts as a buffer solution.

Blood is a mixture that contains iron.

State the function of iron in blood.

It forms complex ion with water $[Fe(H_2O)_6]^{3+}$ and maintains water solution in blood. [1] 0

(d) Blood also contains some proteins.

State **one** effect of toxic metals on proteins.

destroy disulfide bridges (–S–S–bond). [1] 1

Comments for candidate C answer

10a

10b

10c

10d

Candidate E answer

10 A buffer solution is prepared by mixing a weak acid with one of its salts.

For Examiner's Use

(a) Describe what is meant by the term *buffer solution*.

it is the solution where pH value does not change when small amount of alkali or acid is added. [1] 0

(b) The buffer solution contains 0.100 mol dm^{-3} HA and 0.150 mol dm^{-3} NaA.

The acid dissociation constant, K_a, for HA is 1.85 x 10^{-5} mol dm^{-3}.

Calculate the pH of this buffer solution.

$[OH] = K_a \times \frac{[NaA]}{[HA]} = K_a \cdot \frac{0.15}{0.1} = 2.775 \times 10^{-5}$ mol dm^{-3} ✗

$[H] = \frac{10^{-14}}{2.775 \times 10^{-5}} = 3.6 \times 10^{-10}$ ✗ $pH = -\log(3.6 \times 10^{-10}) = 9.44$

pH = 9.44 ✗ [3] 0

(c) Blood acts as a buffer solution.

Blood is a mixture that contains iron.

State the function of iron in blood.

Iron is in structure of haemoglobin and it can react with oxygen, so iron transport oxygen ✓ ✓ [1] 1 1

(d) Blood also contains some proteins.

State **one** effect of toxic metals on proteins.

their damage the protein structure ✓ ✓ [1] 1 1

and the process not work. For example, incompetitive inhibitor on enzyme. Toxic metal like CN can change the active site of enzyme and reaction will not happen.

Comments for candidate E answer

10a

10b

10c

10d